7th lab: AWS Enterprise Networking & Hybrid Connectivity Architecture

Description of the Project

In this lab, you will architect and implement an enterprise-grade hybrid network topology connecting AWS cloud resources with an on-premises data center.

1. **Multi-VPC Infrastructure Provisioning:** You will deploy two distinct VPCs (Production VPC 10.1.0.0/16 and Shared Services VPC 10.2.0.0/16) alongside a simulated On-Premises VPC (172.16.0.0/16).
2. **Centralized Hub-and-Spoke Routing with Transit Gateway:** You will provision an AWS Transit Gateway and attach both AWS VPCs and the on-premises network to replace point-to-point mesh connectivity.
3. **Site-to-Site IPsec VPN Setup:** You will configure a Customer Gateway (CGW), Virtual Private Gateway / TGW VPN Attachment, and establish IPsec tunnels with IPsec BGP/Static routing.
4. **Point-to-Point VPC Peering:** You will establish a VPC Peering connection directly between Production VPC and Shared Services VPC for high-throughput microservice traffic bypassing the TGW where needed.

Step-by-Step Implementation Guide

Step 1: Network Topology & VPC Infrastructure Setup

Create three distinct network environments with non-overlapping CIDR blocks:

- Navigate to **VPC Console ➔** Create the following VPCs:
  - **Production VPC:** CIDR 10.1.0.0/16 | 2 Private Subnets (10.1.1.0/24, 10.1.2.0/24).
  - **Shared Services VPC:** CIDR 10.2.0.0/16 | 2 Private Subnets (10.2.1.0/24, 10.2.2.0/24).
  - **Simulated On-Premises VPC:** CIDR 172.16.0.0/16 | 1 Public Subnet (172.16.1.0/24) hosting an EC2 router/DNS server.
- Enable **DNS Hostnames** and **DNS Resolution** across all three VPCs.

Step 2: Transit Gateway (TGW) & VPC Attachments

- Navigate to **VPC Console ➔ Transit Gateways ➔** Click **Create Transit Gateway**:
  - **Name:** enterprise-tgw | **Amazon Side ASN:** 64512.
  - **Auto Accept Shared Attachments:** Enabled.

- Navigate to **Transit Gateway Attachments ➔** Create Attachments:
  - **Attachment 1:** Type = VPC | Select Production VPC | Attach private subnets.
  - **Attachment 2:** Type = VPC | Select Shared Services VPC | Attach private subnets.
- Configure **TGW Route Tables**:
  - Create prod-tgw-rt and shared-tgw-rt to isolate or allow cross-spoke communication.
  - Add routes pointing 10.2.0.0/16 traffic to Shared Services attachment and 10.1.0.0/16 to Production attachment.

Step 3: Establish AWS Site-to-Site VPN to On-Premises

- Navigate to **Virtual Private Network (VPN) ➔ Customer Gateways**:
  - **Name:** onprem-cgw | **IP Address:** Public IP of simulated On-Premises EC2 router.
- Navigate to **Site-to-Site VPN Connections ➔** Click **Create VPN Connection**:
  - **Name:** tgw-onprem-vpn.
  - **Target Gateway:** Transit Gateway (enterprise-tgw).
  - **Customer Gateway:** Select onprem-cgw.
  - **Routing Options:** Static (Add static route 172.16.0.0/16).
- Download the VPN configuration file and apply strongswan / IPsec configuration on the On-Premises router instance. Verify Tunnel 1 status turns **UP**.

Step 4: Configure VPC Peering Connection

- Navigate to **VPC Console ➔ Peering Connections ➔** Click **Create Peering Connection**:
  - **Name:** prod-to-shared-peering.
  - **Requester:** Production VPC (10.1.0.0/16).
  - **Accepter:** Shared Services VPC (10.2.0.0/16).
- Accept the peering request.
- Update Route Tables:
  - In Production VPC Route Table: Add route 10.2.0.0/16 ➔ Target: pcx-xxxxxx.
  - In Shared Services VPC Route Table: Add route 10.1.0.0/16 ➔ Target: pcx-xxxxxx.

Step 5: Hybrid DNS Resolution via Route 53 Resolver

- **Private Hosted Zone Creation:** Create Private Hosted Zone cloud.aws.internal in Route 53 and associate with Production & Shared Services VPCs.
- **Route 53 Inbound Resolver Endpoint:**
  - Navigate to **Route 53 Console ➔ Inbound Endpoints ➔** Create Inbound Endpoint in Shared Services VPC across 2 Availability Zones.
  - Allows on-premises workloads to query cloud.aws.internal records via Inbound IP addresses.
- **Route 53 Outbound Resolver Endpoint & Forwarding Rules:**
  - Create **Outbound Endpoint** in Shared Services VPC across 2 Availability Zones.
  - Create **Resolver Rule**: Rule Type = Forward | Domain Name = corp.internal | Target IP = On-Premises DNS Server IP (172.16.1.53).
  - Associate rule with Production and Shared Services VPCs.

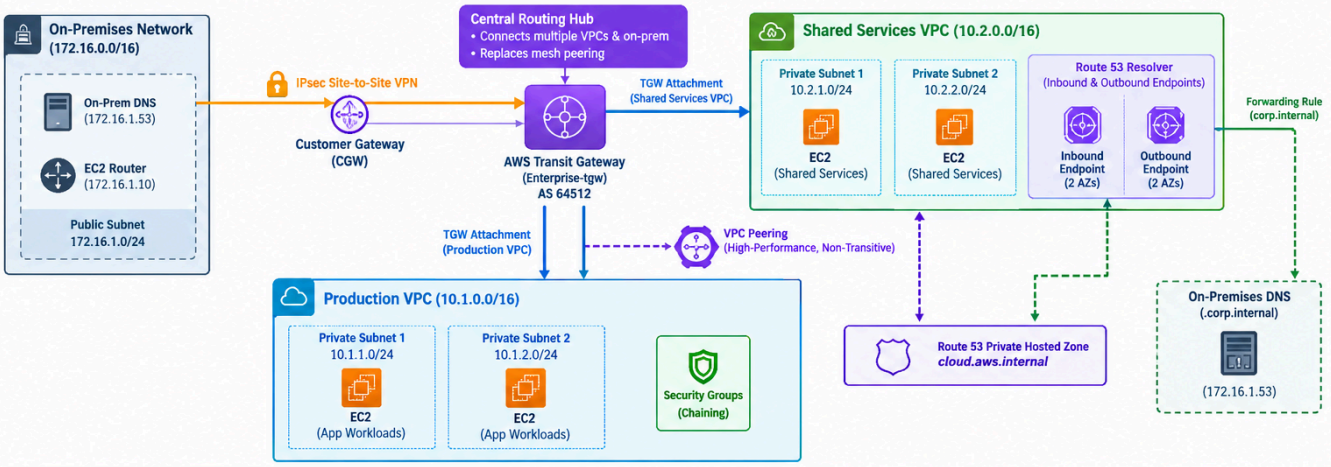
Step 6: Testing & Hybrid Connectivity Validation

- **Inter-VPC Ping & Curl Test:** SSH into an EC2 instance in Production VPC and run ping / curl to Shared Services private IP over VPC Peering.
- **On-Premises VPN Verification:** Ping from Production VPC (10.1.1.X) to On-Premises instance (172.16.1.X) through Transit Gateway over the VPN tunnel.
- **Hybrid DNS Resolution Test:**
    - From AWS EC2 instance: Execute dig app.corp.internal and verify it resolves via Route 53 Outbound Endpoint.
    - From On-Premises machine: Execute dig db.cloud.aws.internal @<INBOUND-RESOLVER-IP> and verify resolution of cloud private IPs.

AWS Enterprise Networking & Hybrid Connectivity Architecture

Transit Gateway | Site-to-Site VPN | VPC Peering | Route 53 Resolver | Hybrid DNS | Security Groups | Route Table Governance

Lab 7
Estimated Duration: 1.5 – 2 Hours



Component Overview

Component	AWS Service / Feature	Architectural Purpose
Central Routing Hub	Transit Gateway (TGW)	Interconnects multiple VPCs and on-premises (central cloud router).
Hybrid Connectivity Tier	Site-to-Site VPN	Encrypted IPsec tunnels between on-prem and TGW.
High-Performance Inter-VPC Link	VPC Peering	Low-latency, non-transitive link between Production and Shared Services VPCs.
Hybrid DNS Resolution Tier	Route 53 Resolver (Endpoints & Rules)	Bi-directional DNS resolution between AWS and on-premises.
Network Security Tier	TGW Route Tables & Security Groups	Network segmentation and restrictive access control.

Key Route Tables

TGW Route Tables

Route	Target
10.1.0.0/16	→ Production Attachment
10.2.0.0/16	→ Shared Services Attachment

VPC Route Tables

Route	Target
10.2.0.0/16	→ VPC Peering (from Production)
10.1.0.0/16	→ VPC Peering (from Shared Services)

Security Groups (Chaining)

App SG (Production)
Allow 10.2.0.0/16 (Shared Services)

→

Shared Services SG (Shared Services)
Allow 10.1.0.0/16 (Production)

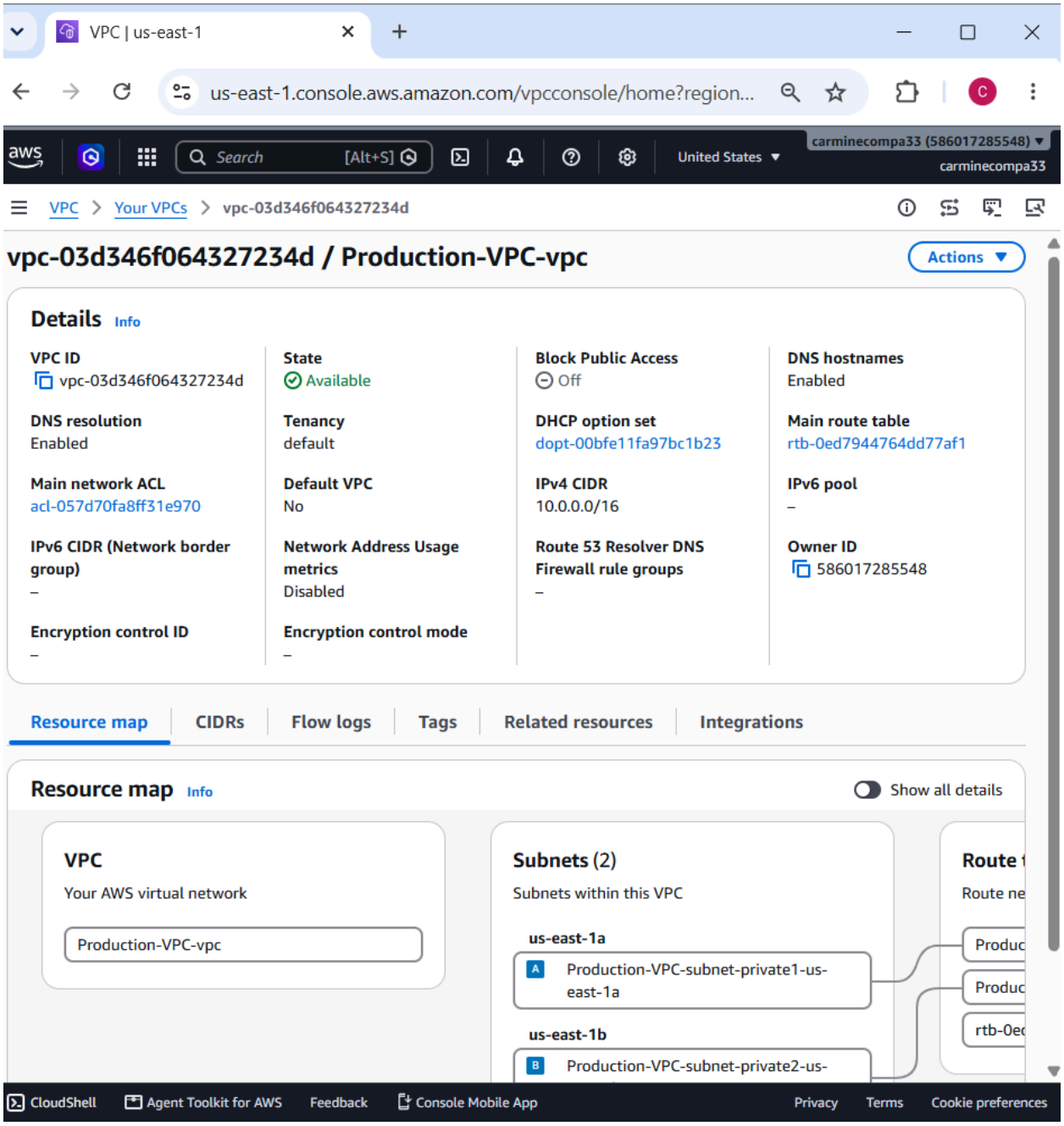
- Restrict inbound/outbound rules
- Allow only required ports (e.g., 80, 443, 22)
- Use SG chaining for layered security

Implementation Steps

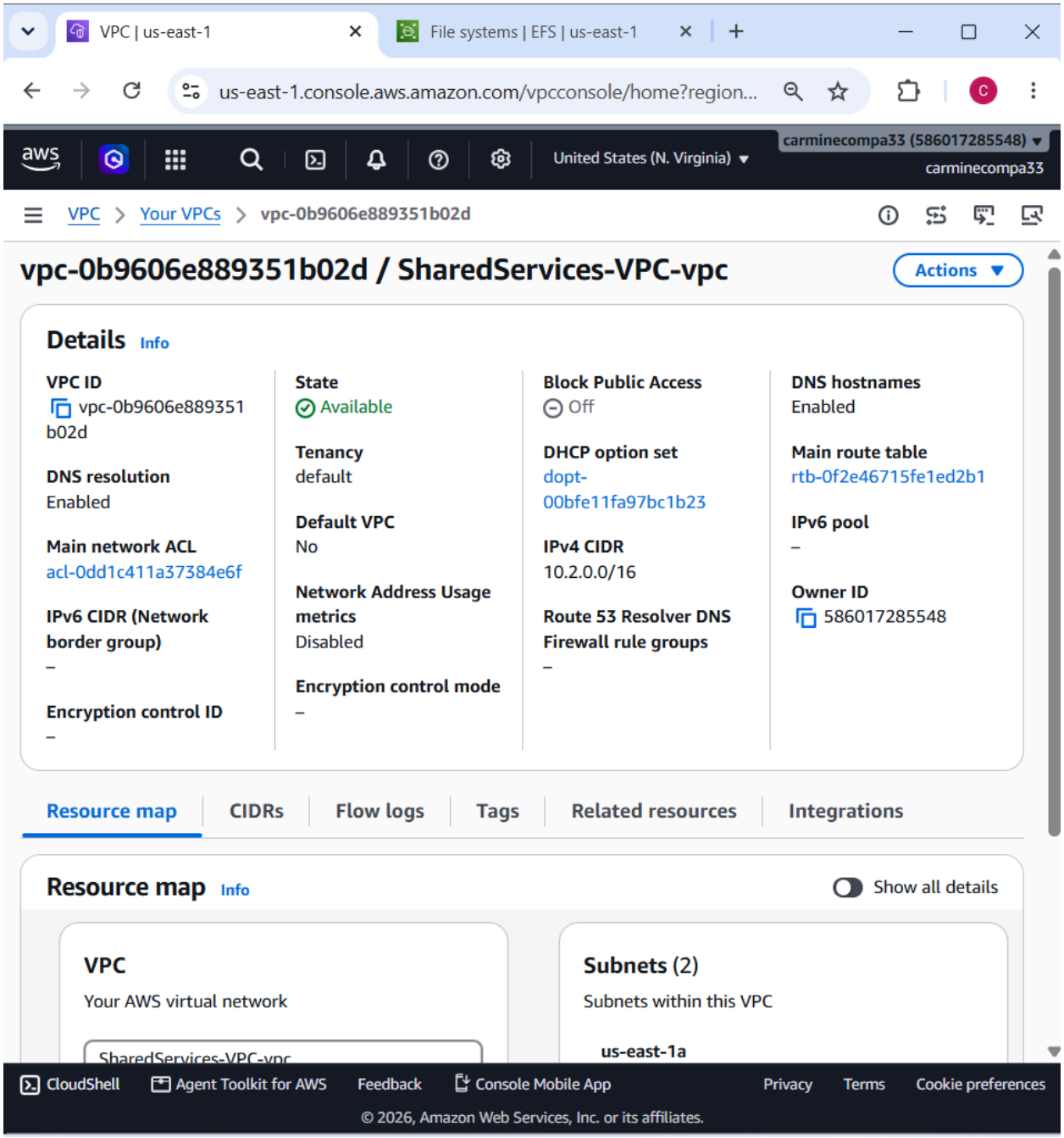
1. Create VPCs (Prod 10.1.0.0/16, Shared 10.2.0.0/16, On-Prem 172.16.0.0/16) + subnets + DNS settings
2. Create TGW + attachments + route tables
3. Configure Site-to-Site VPN (CGW, VPN, static/BGP)
4. Establish VPC Peering (Prod ↔ Shared) + route tables
5. Set up Route 53 Resolver (Hosted Zone, Inbound/Outbound, forwarding rules)
6. Test connectivity (ping, curl, dig)

End Result: Secure, scalable, and highly available hybrid network with centralized routing, optimized connectivity, and seamless DNS resolution between AWS and on-premises.

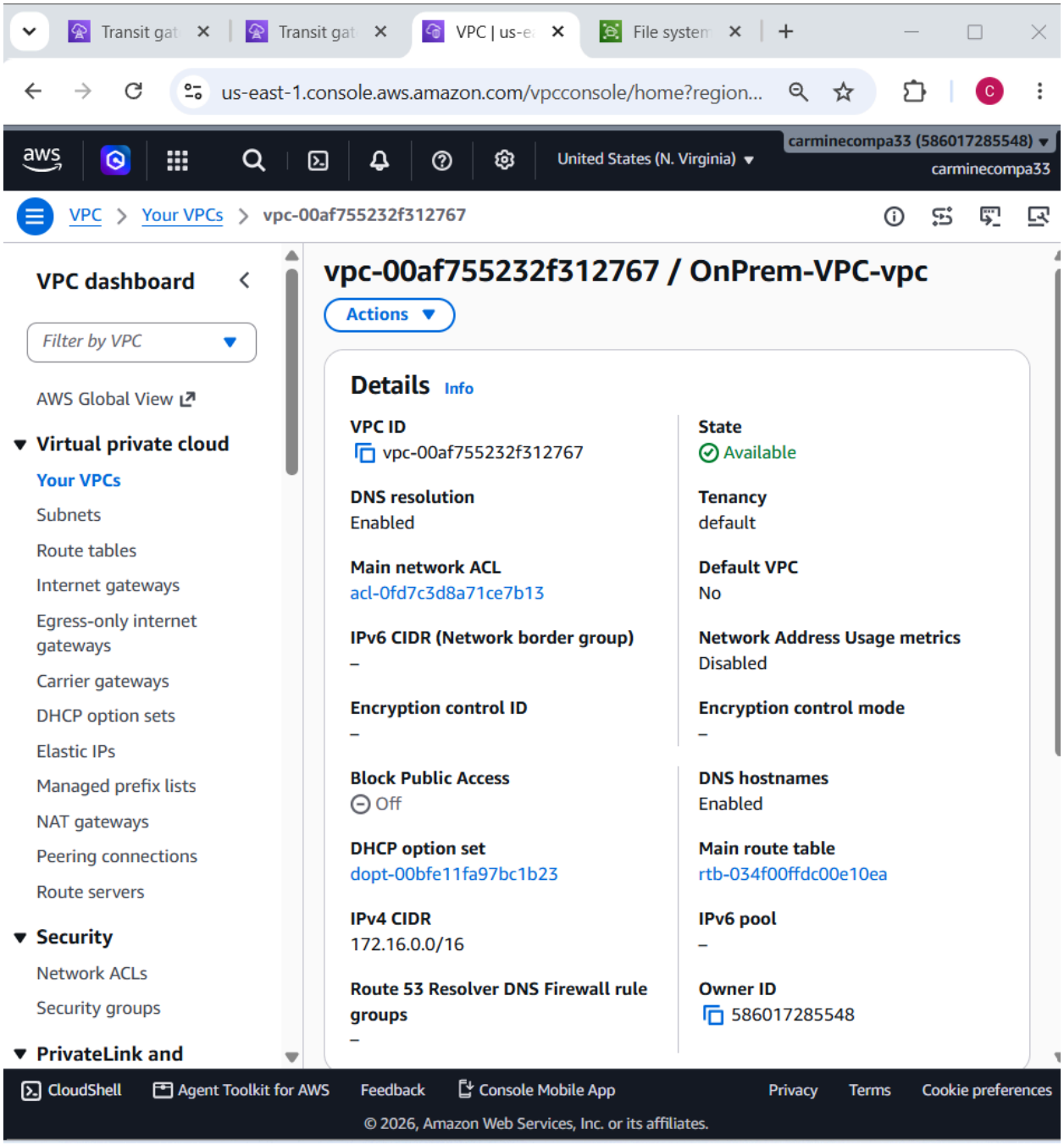
Created Production VPC 10.1.0.0/16 with 2 private subnets and 0 public subnets



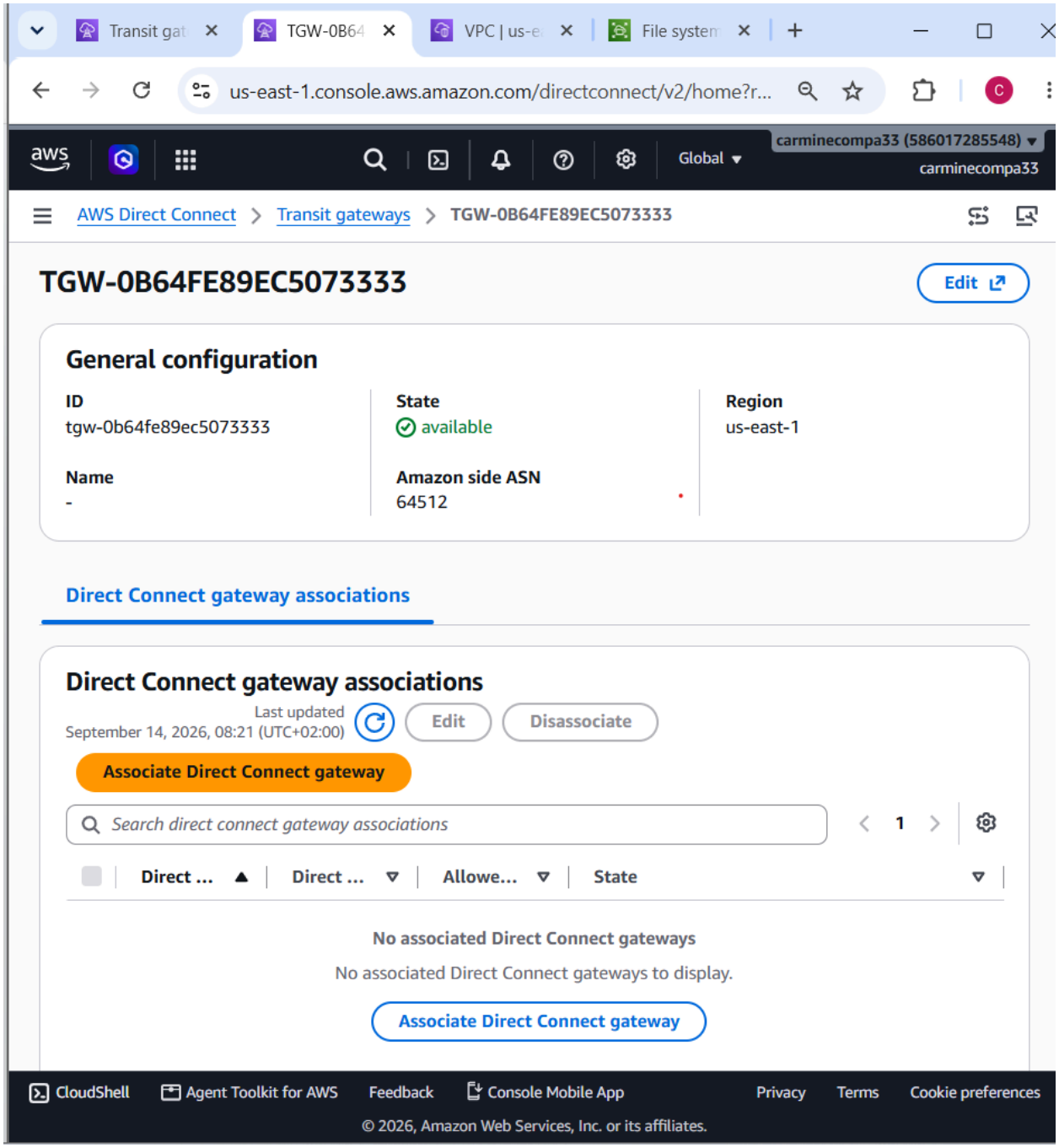
Created SharedNetworkVPC with 2 private subnets and 0 public subnets



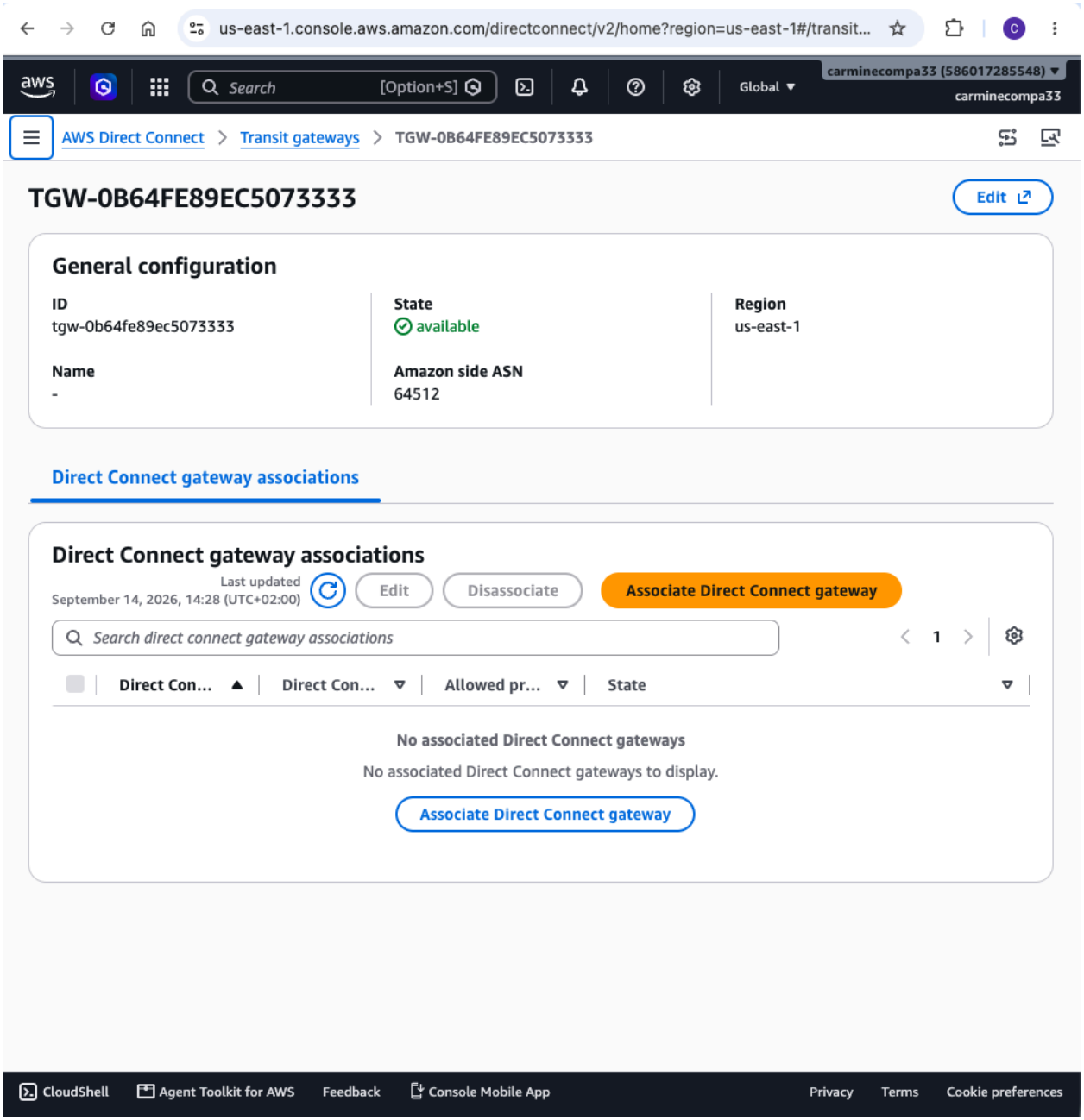
Created simulated OnPrem VPC with 0 private subnets and 1 public subnets



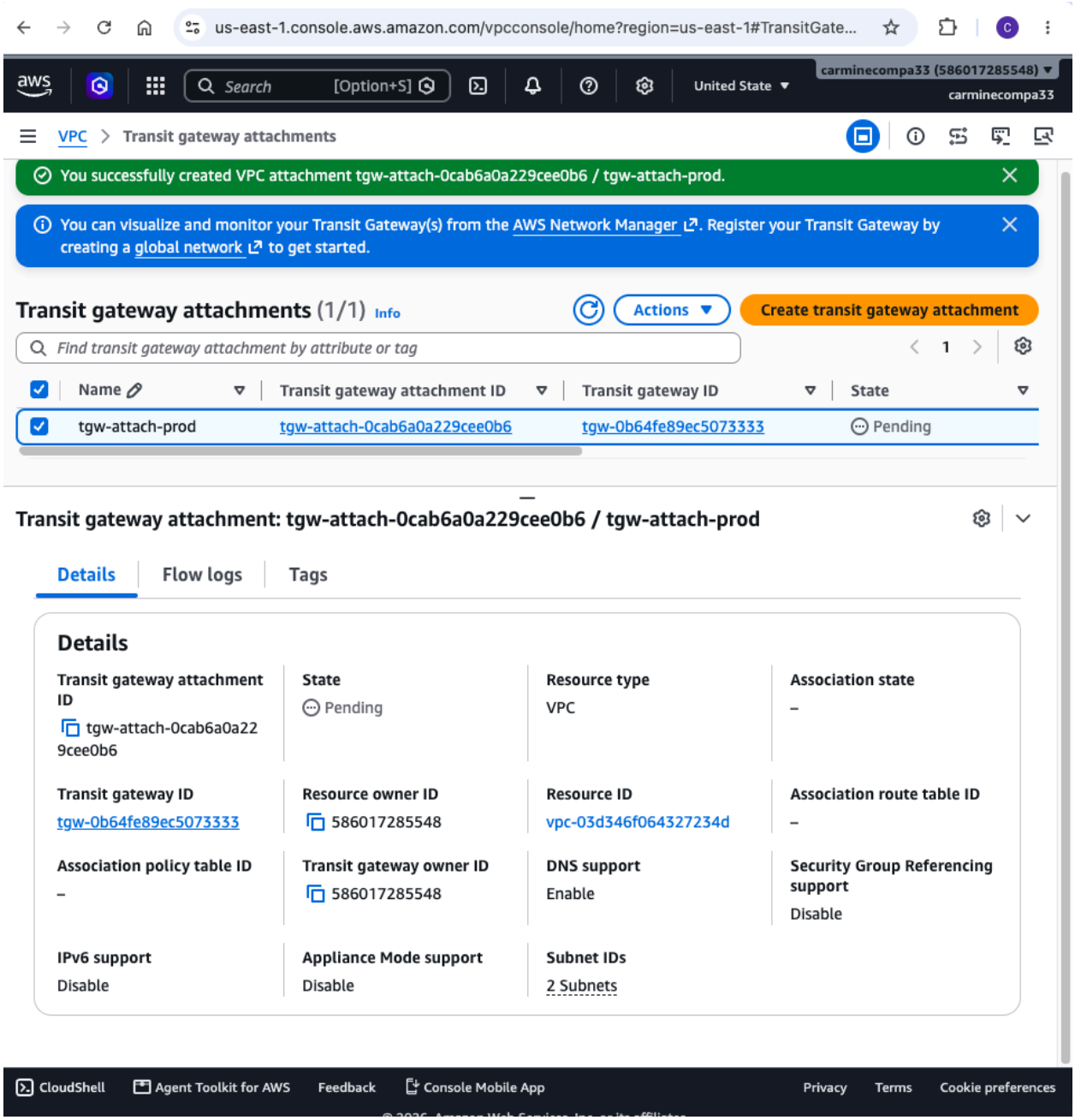
Created Transit Gateway



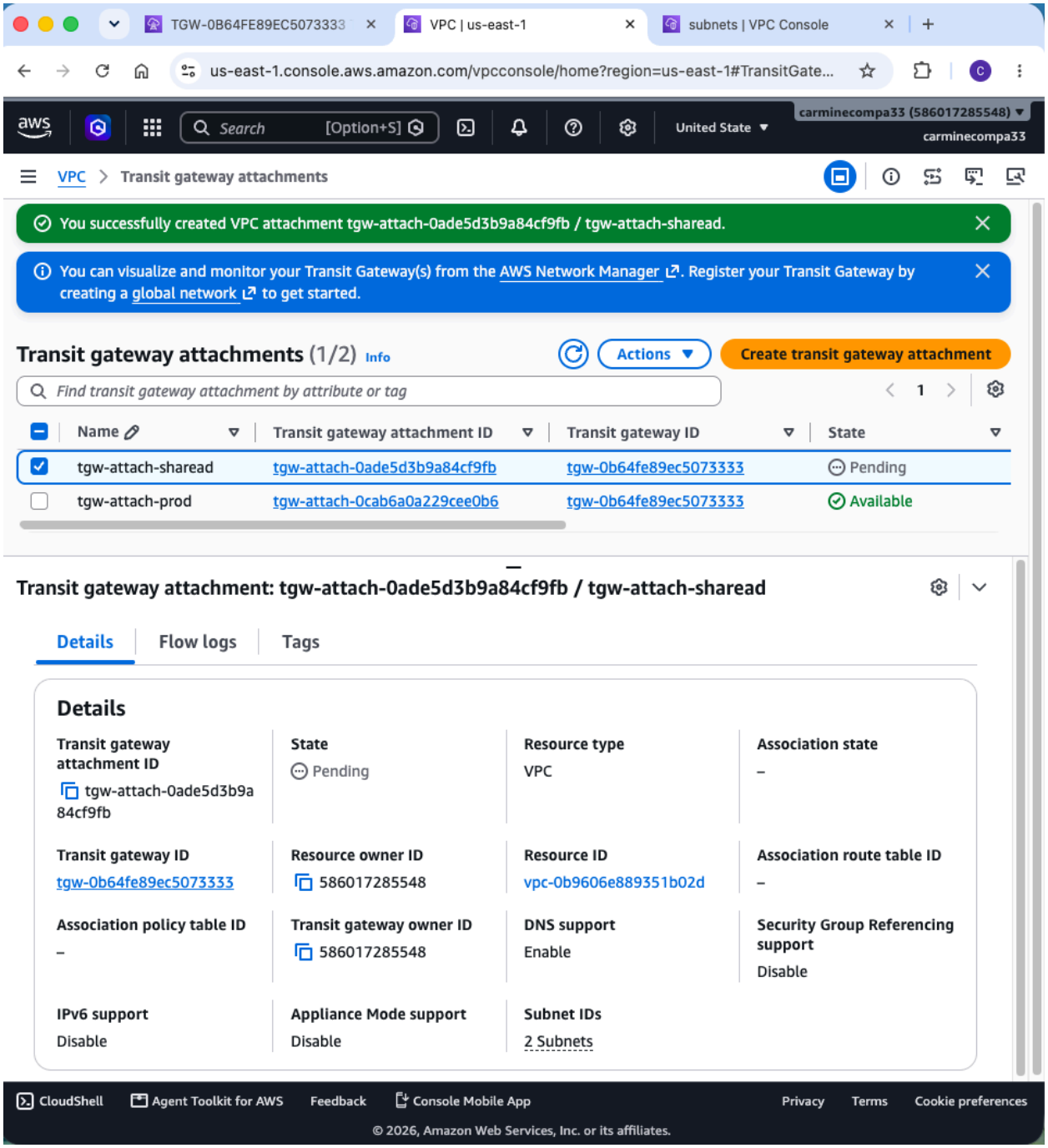
Created Transit Gateway



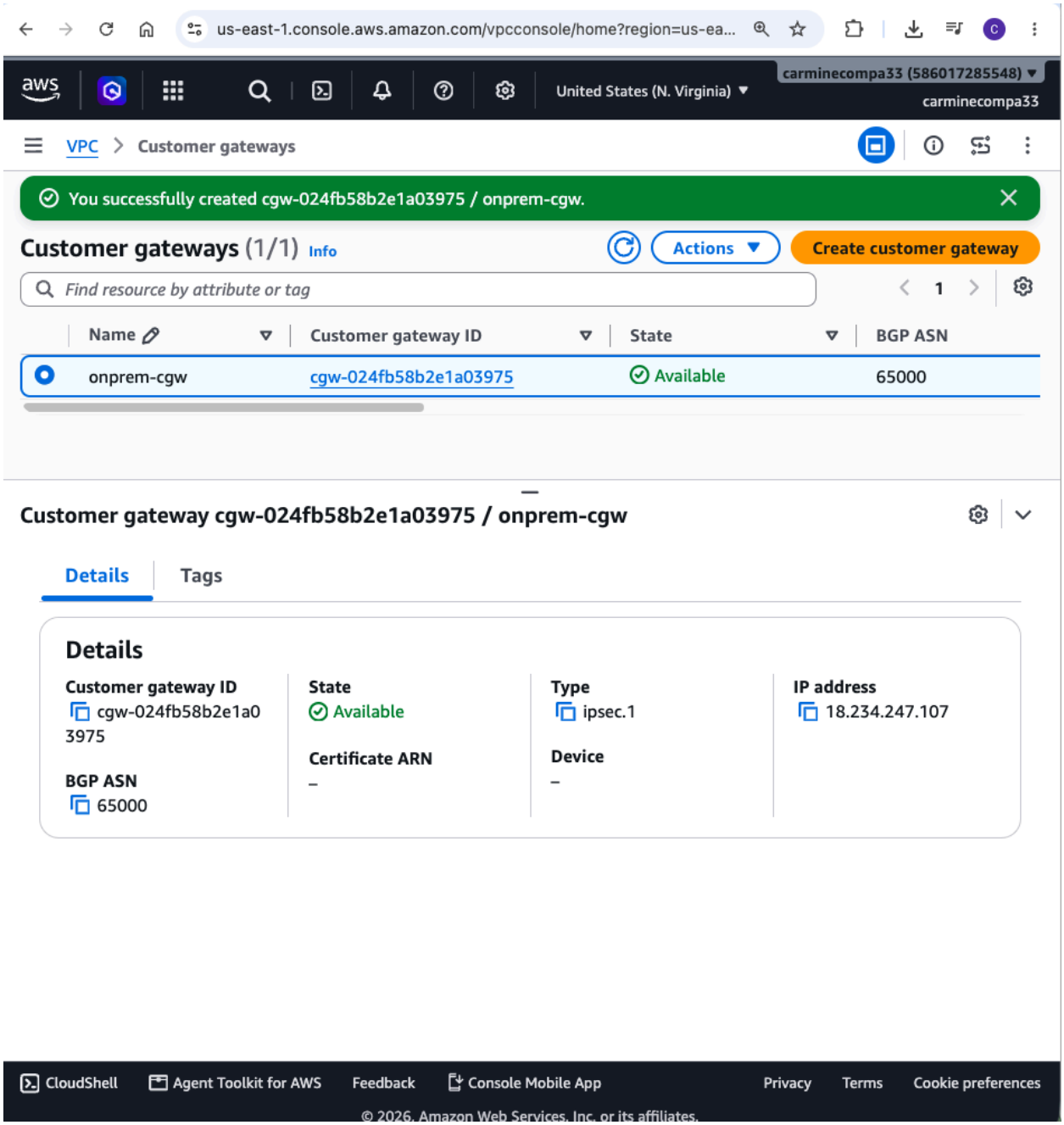
I created Transit Gateway attachment for Production-VPC
I connected the two private subnets of Production-VPC



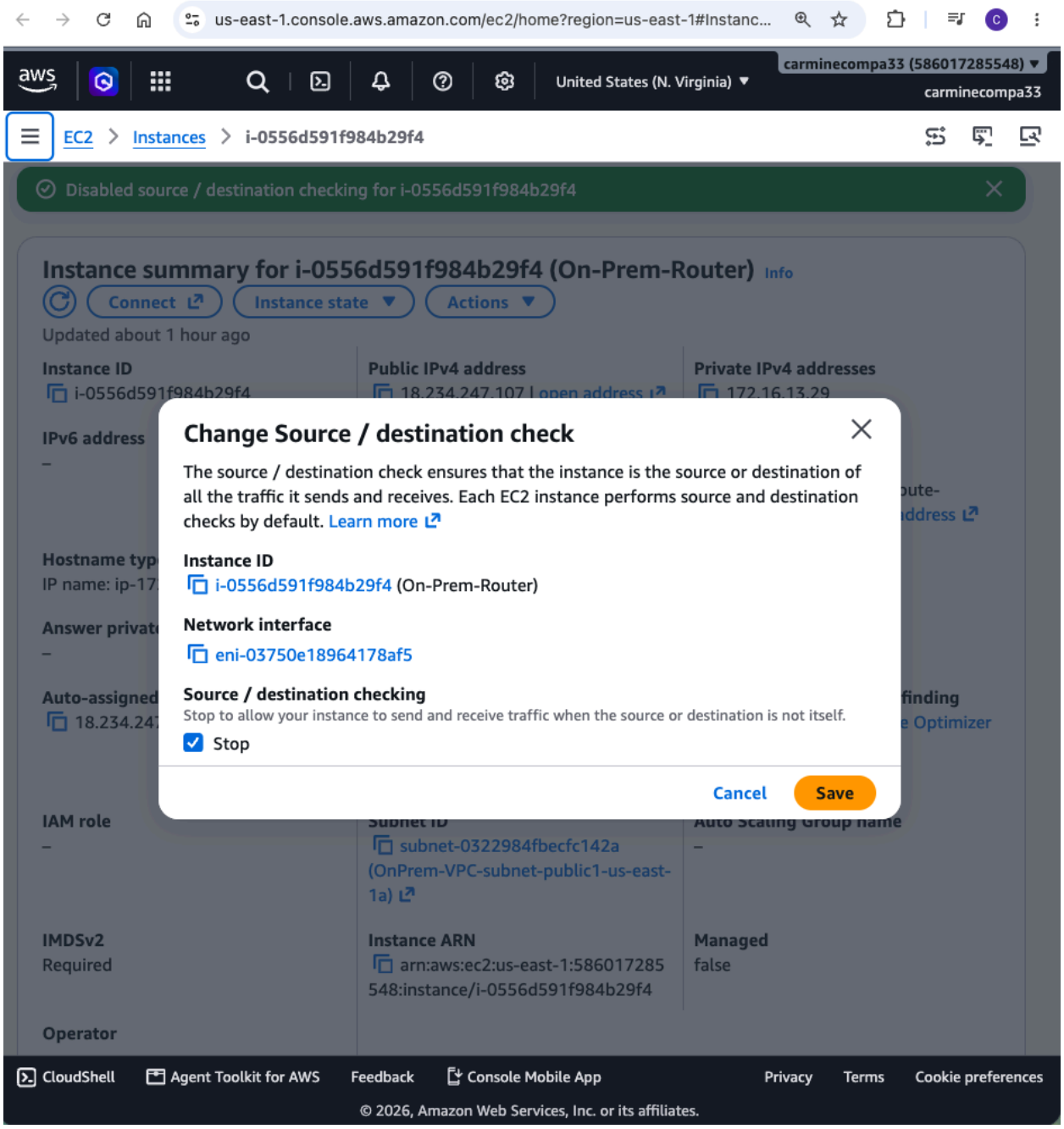
Created TransitGateway attachment to SharedServiceVPC
Attached the two private subnets to the Transit Gateway attachment



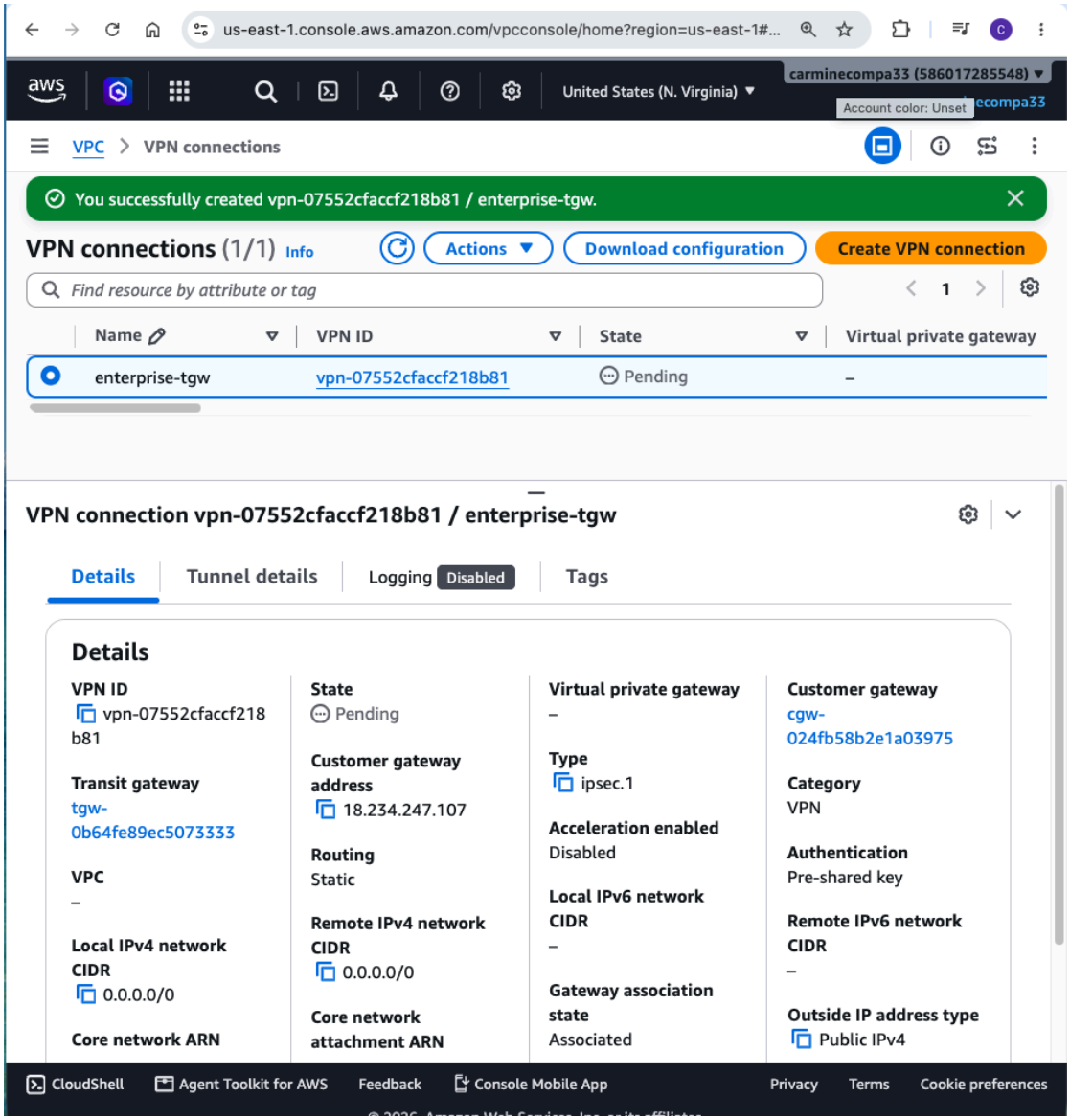
Created Customer Gateway with the OnPrem IP
Customer Gateway stays between OnPrem Network and Customer Gateway



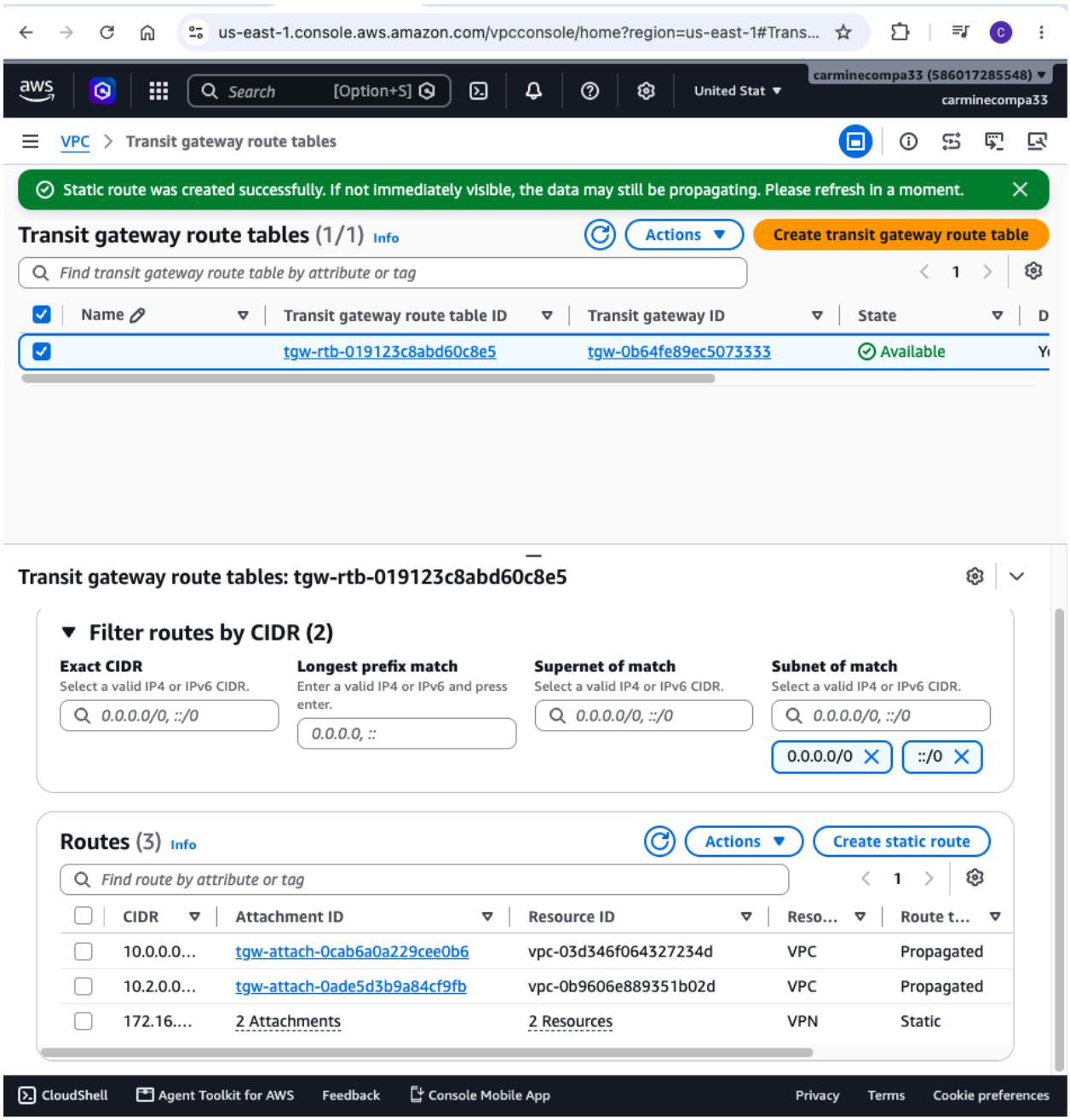
Stopped source/destination check: this allows us to adopt the EC2 as a router



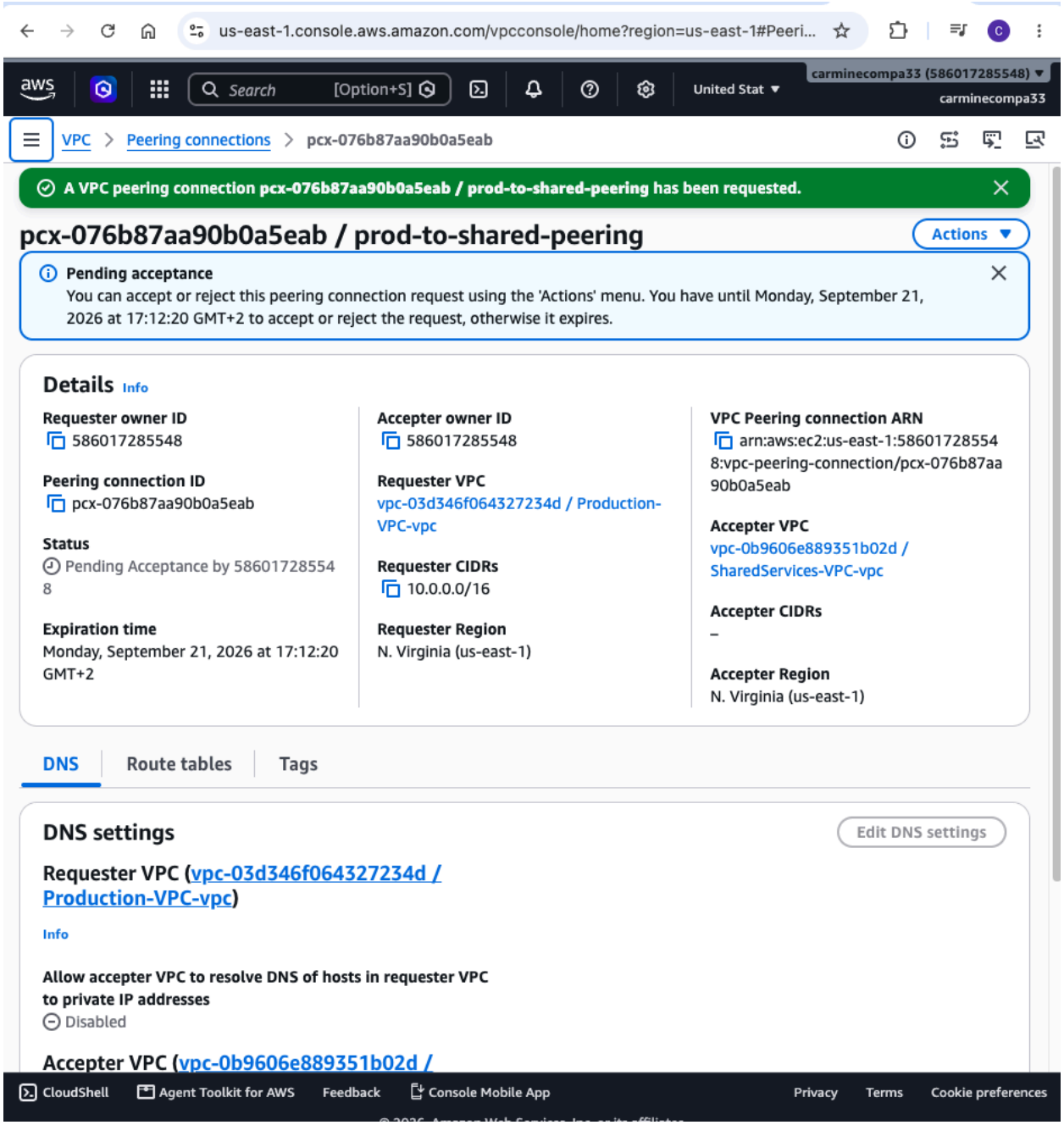
Created VPC connection between Customer Gateway and Transit Gateway



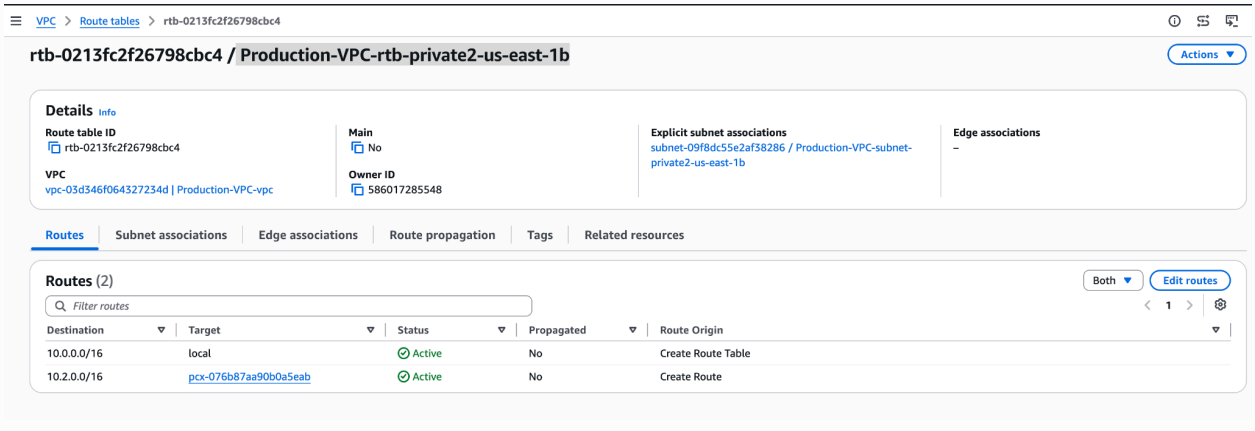
Created static route in Transit Gateway route table
Destination CIDR: 172.16.0.0/16 >> network of the simulated OnPrem Network
Target attachment >> points directly to the IPsec Site-to-Site VPC connection



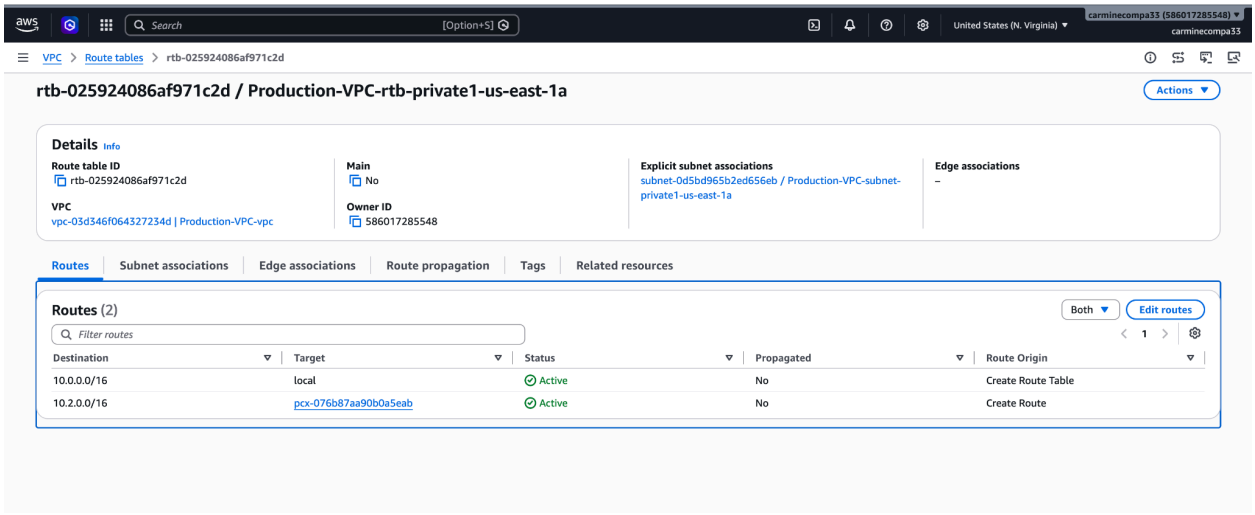
Created a VPC peering connection between prod VPC and shared services VPC



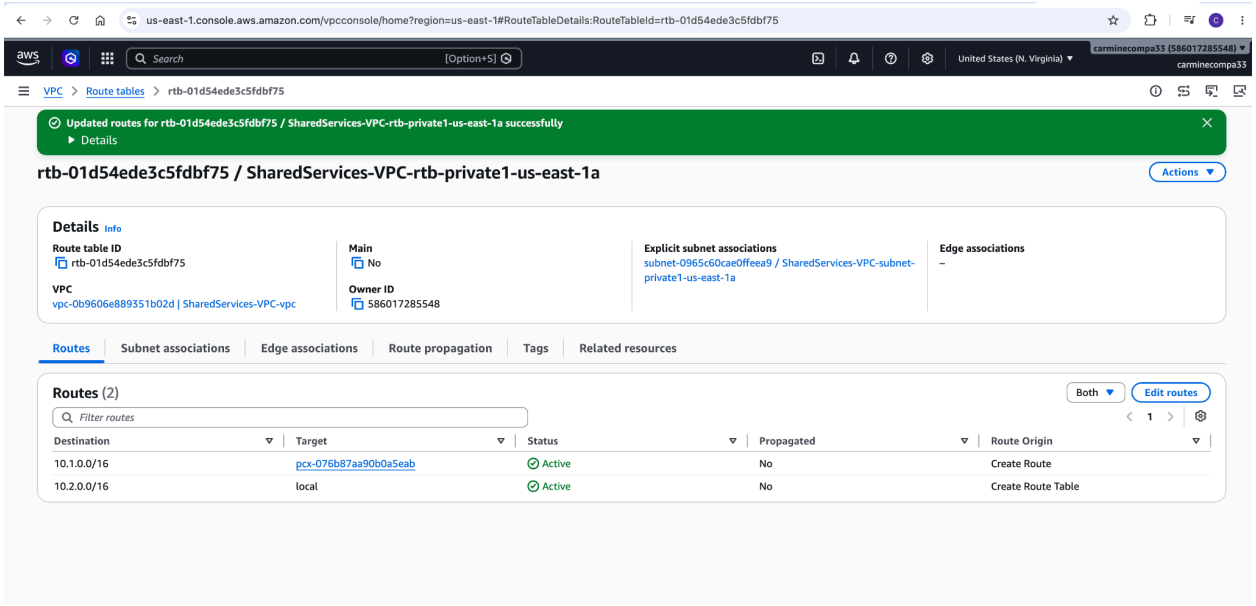
Added route in VPC route table Production-VPC-rtb-private2-us-east-1b of the peering connection with Shared Service VPC



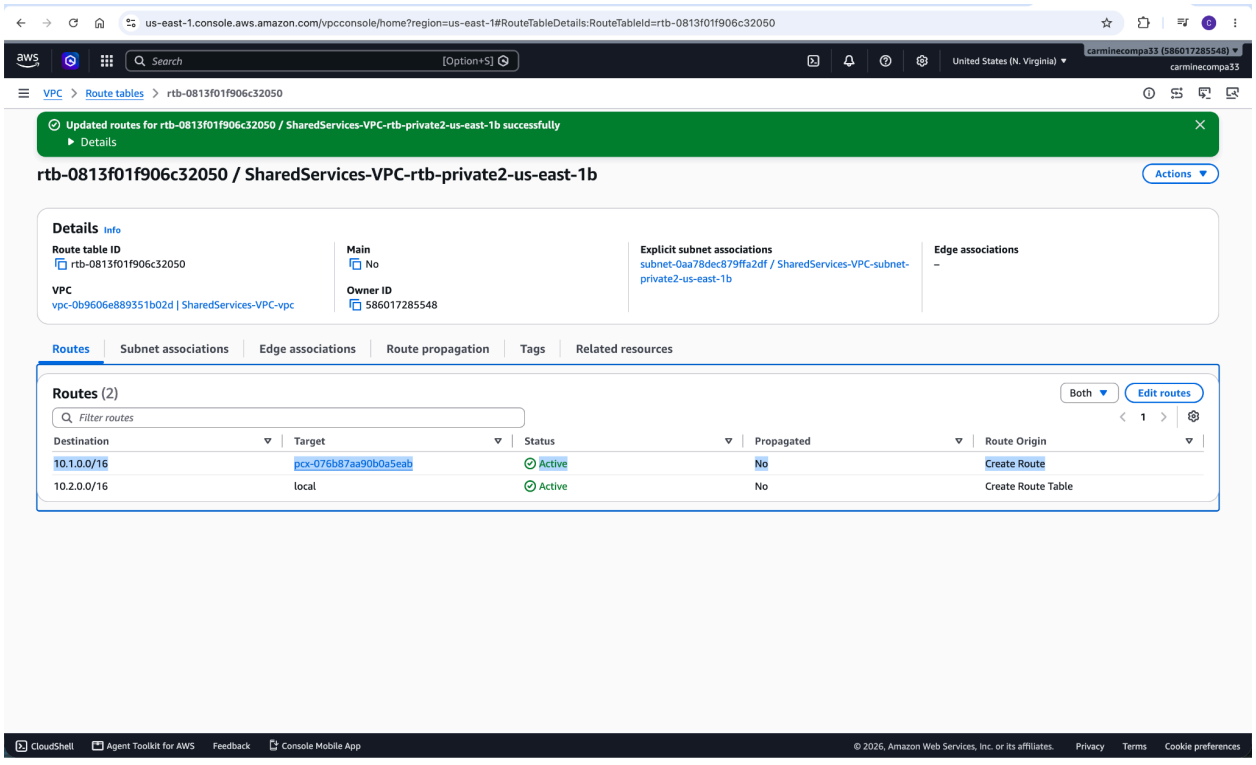
Added route in VPC route table Production-VPC-rtb-private1-us-east-1a of the peering connection with the Shared Services VPC



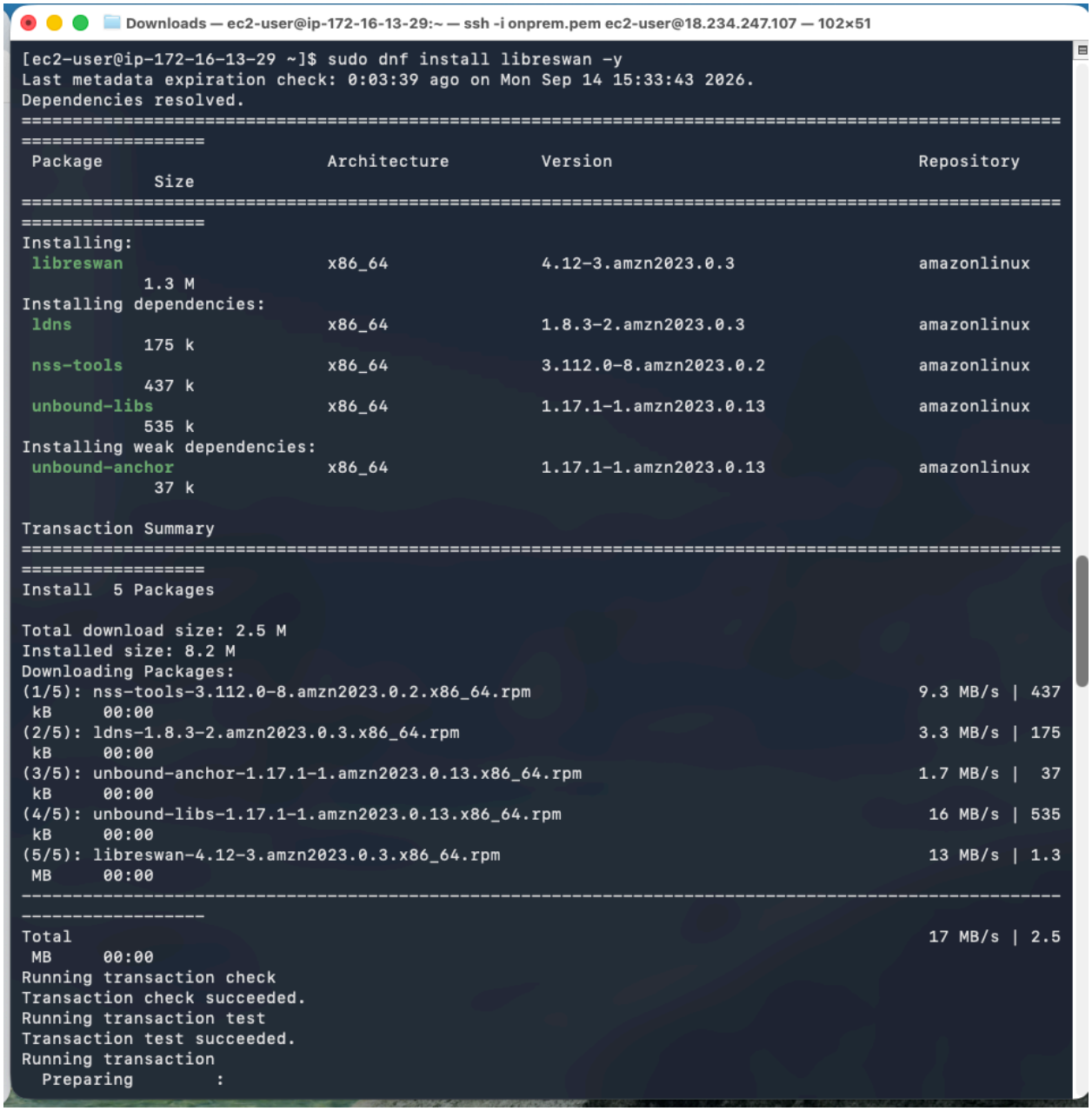
Added from ShareServices VPC from us-east-1 a route to destination IP of Prod PVC



Added route from SharedVPC us-east-1b to Prod VPC CIDR IP 10.1.0.0/16



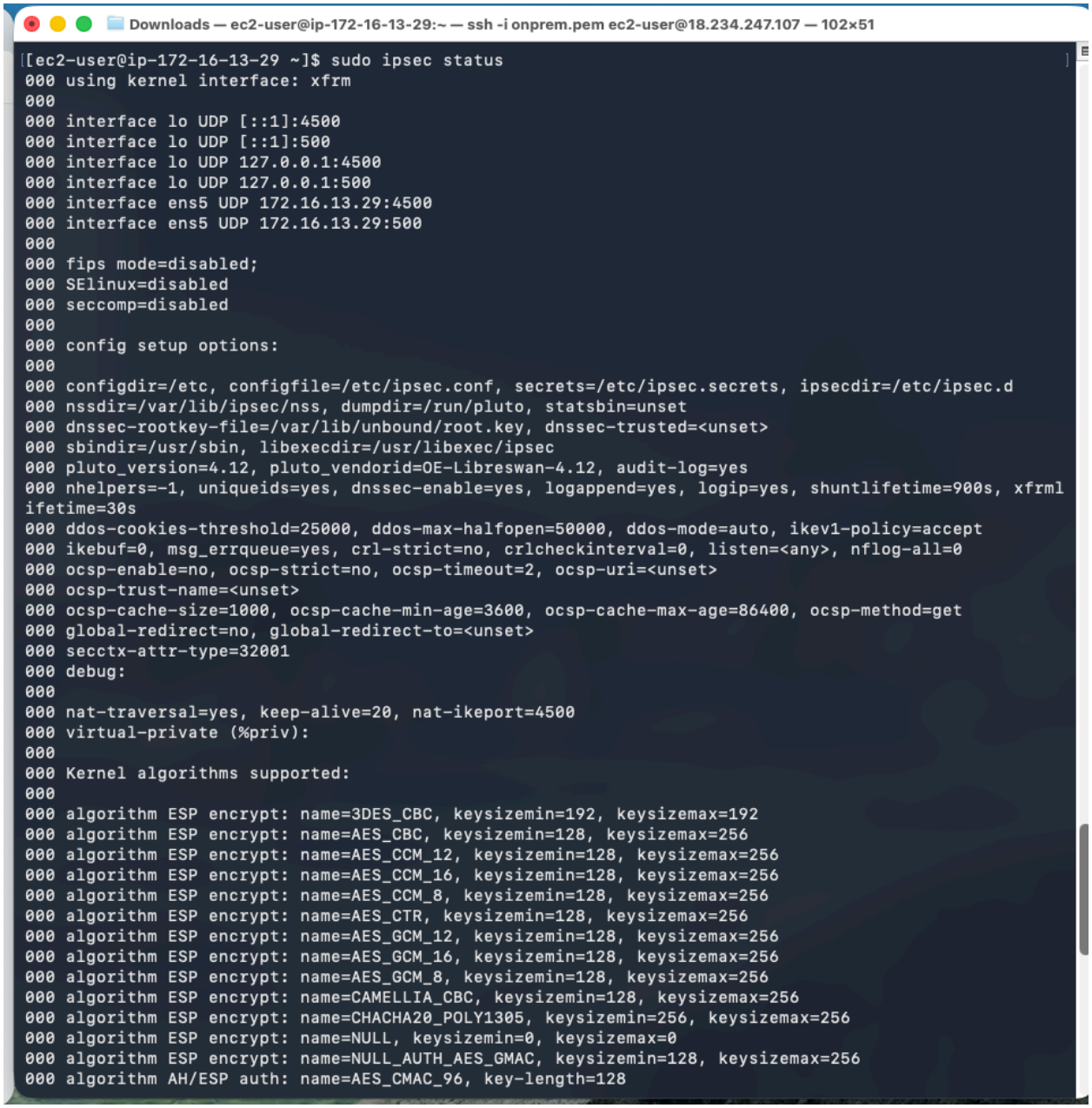
Installed libreswan on the EC2 instance (simulated external OnPrem server):



Enabled ipsec
Enabled net.ipv4.ip_forward
Added Site-to-Site VPN configuration on /etc/ipsec.d/aws-vpc.secrets
Restarted ipsec



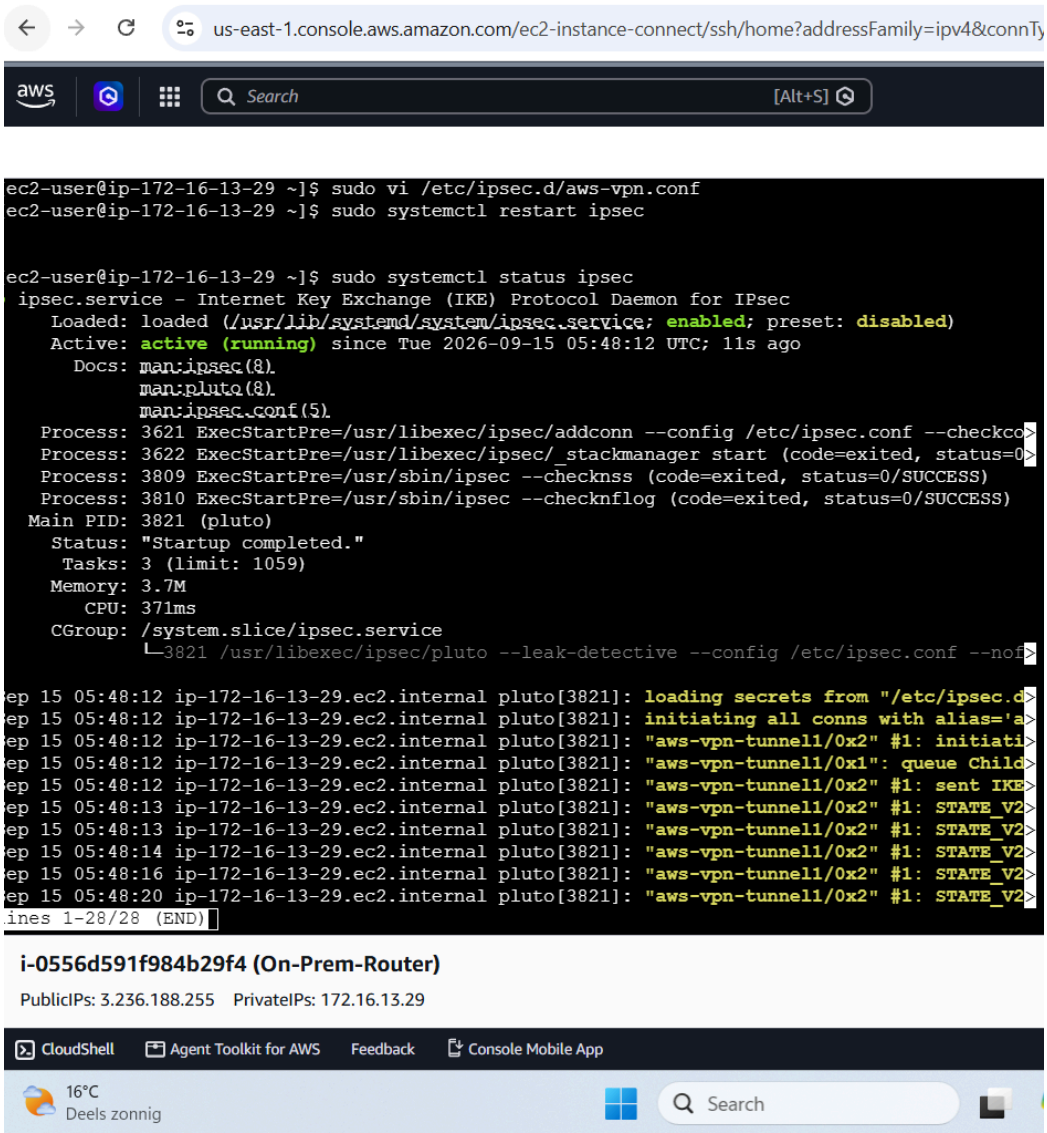
Checked ipsec status >> but it shows Total IPsec connections loaded 0, active 0



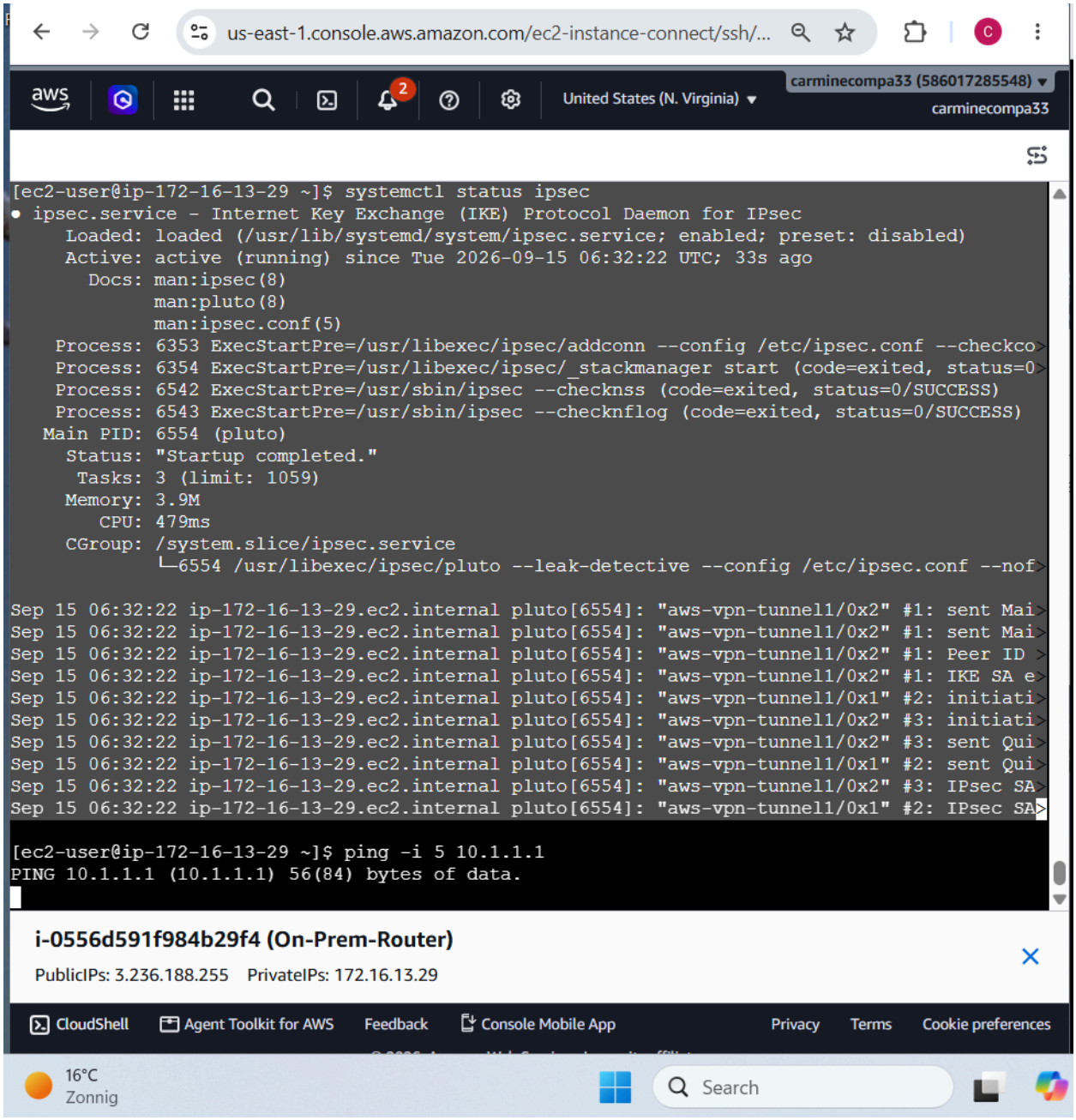
Explored the errors of sudo journalctl -u ipsec
Found the below error:

"aws-vpn-tunnel1/0x1": failed to add connection: AEAD ESP encryption algorithm AES_GCM_16 must have 'NONE' as the integrity algorithm

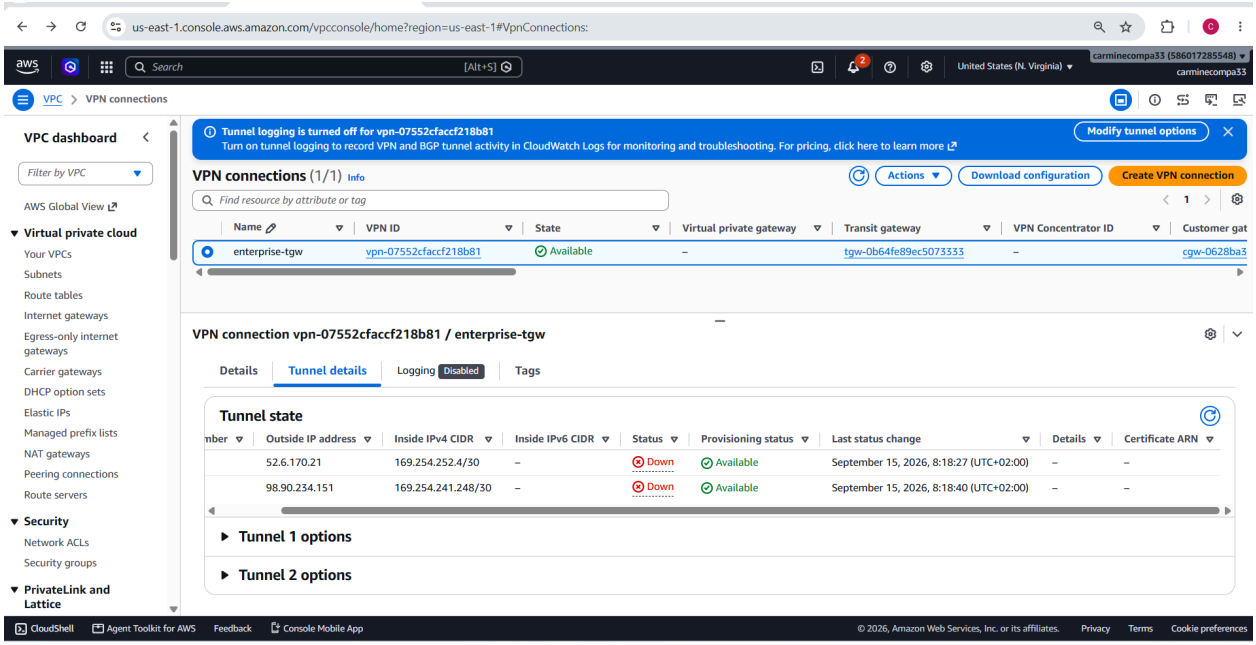
And therefore modified the file /etc/ipsec.d/aws-vpc.conf by setting:
ike=aes256-sha256:modp2048
phase2alg=aes256-sha256;modp2048



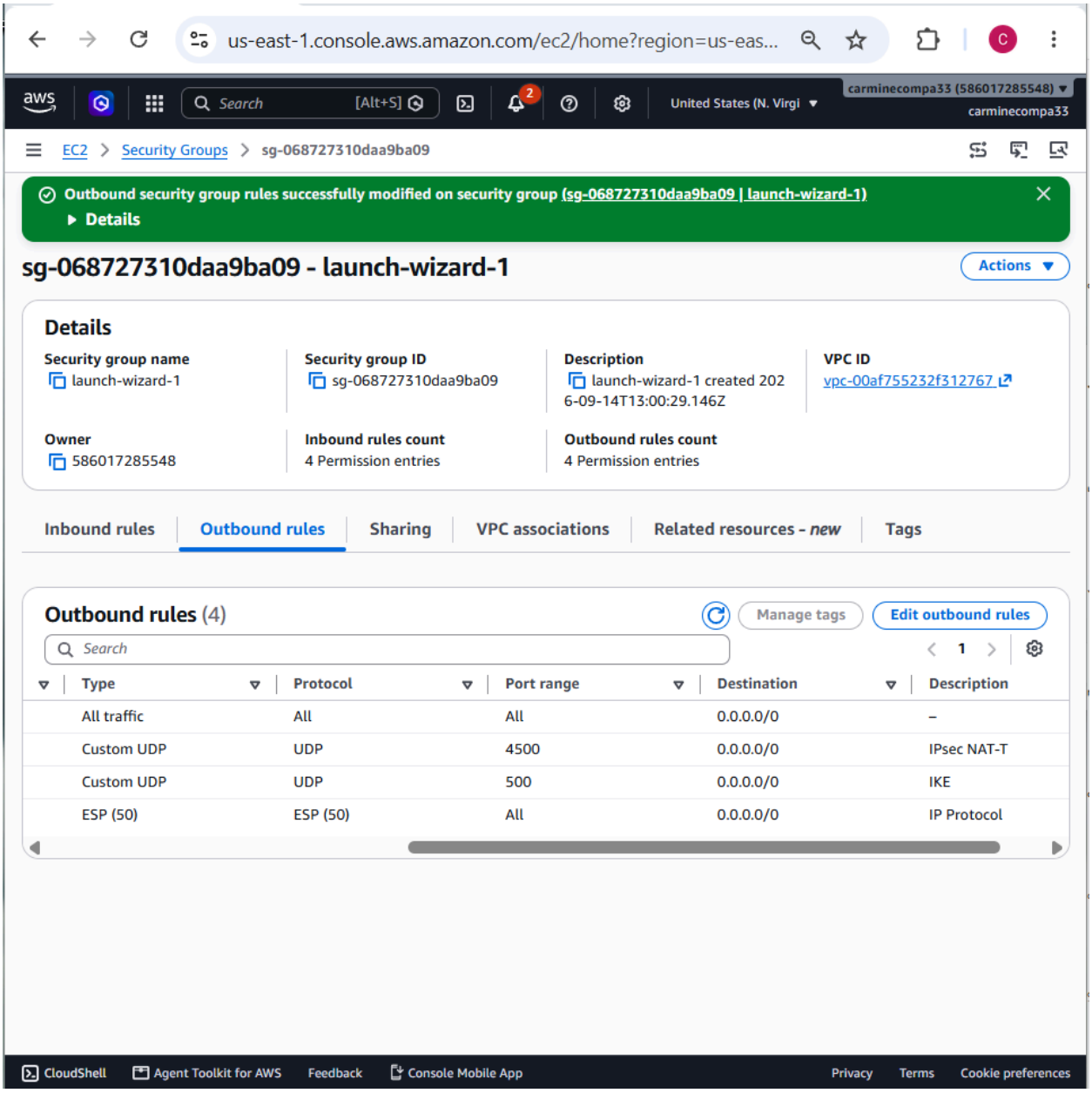
Successfully generated ipsec locally

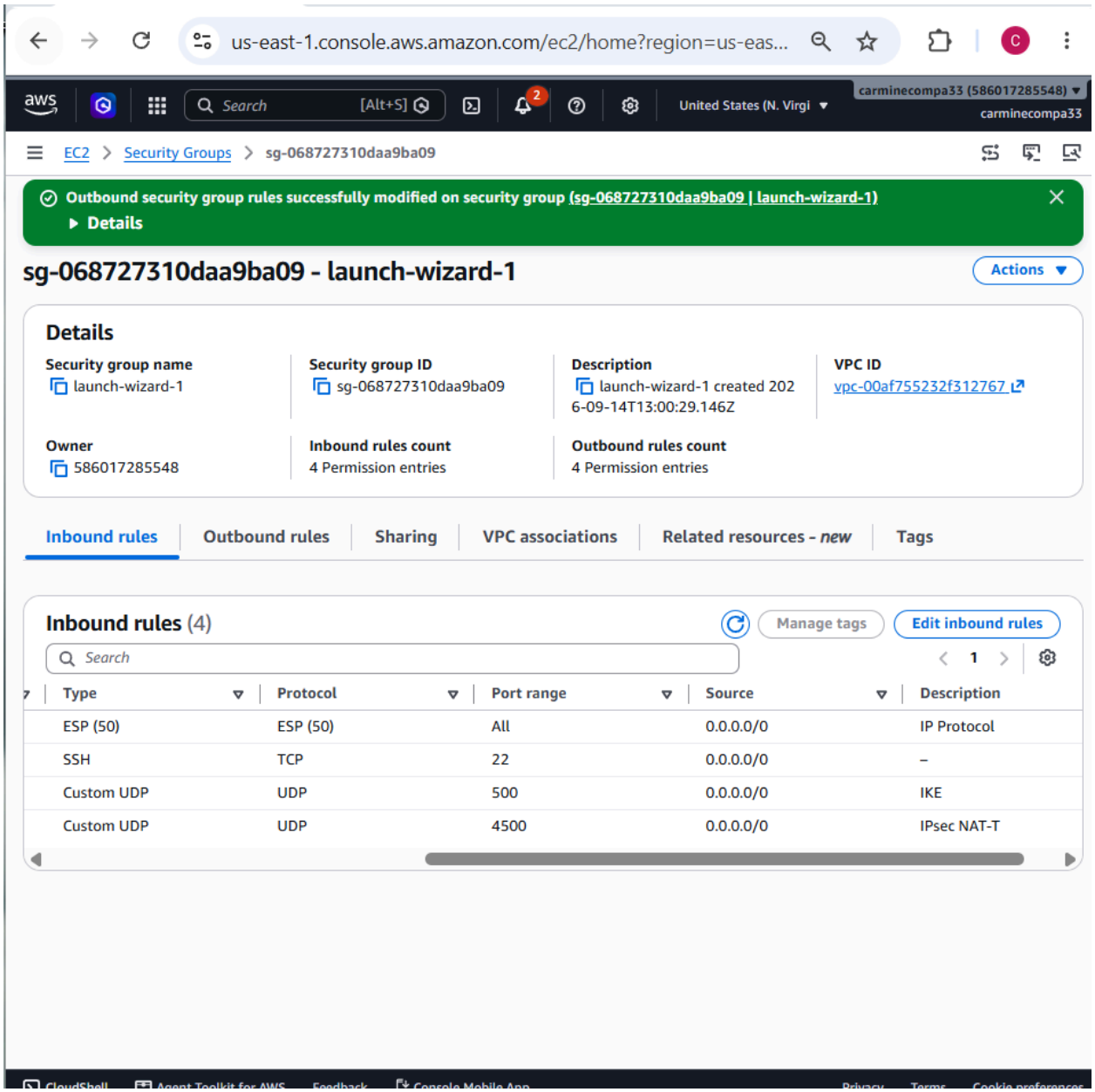


However the tunnel 1 that I configured it is still down in AWS

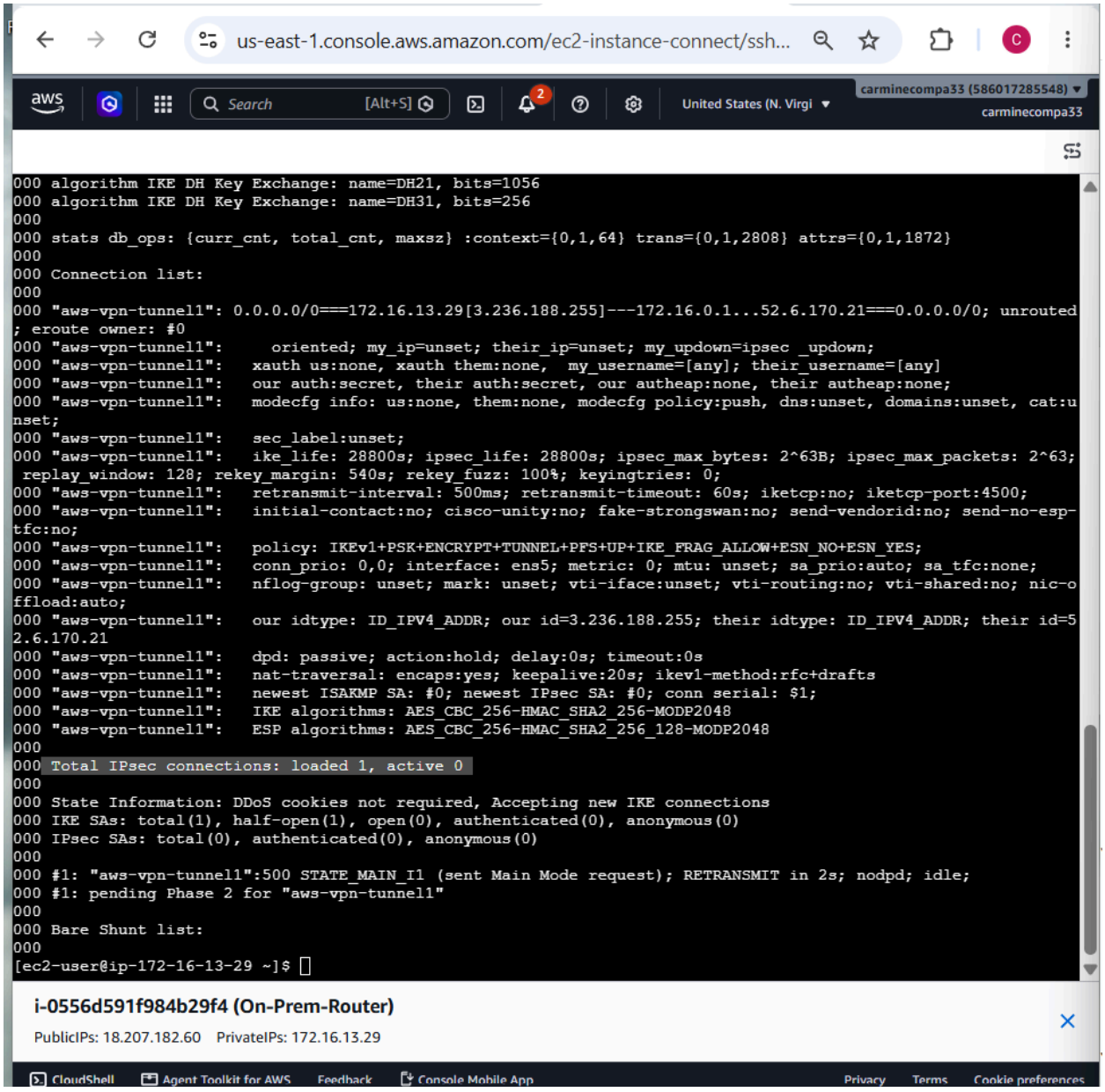


Added outbound rules
And added ESP (Protocol 50) to the rules

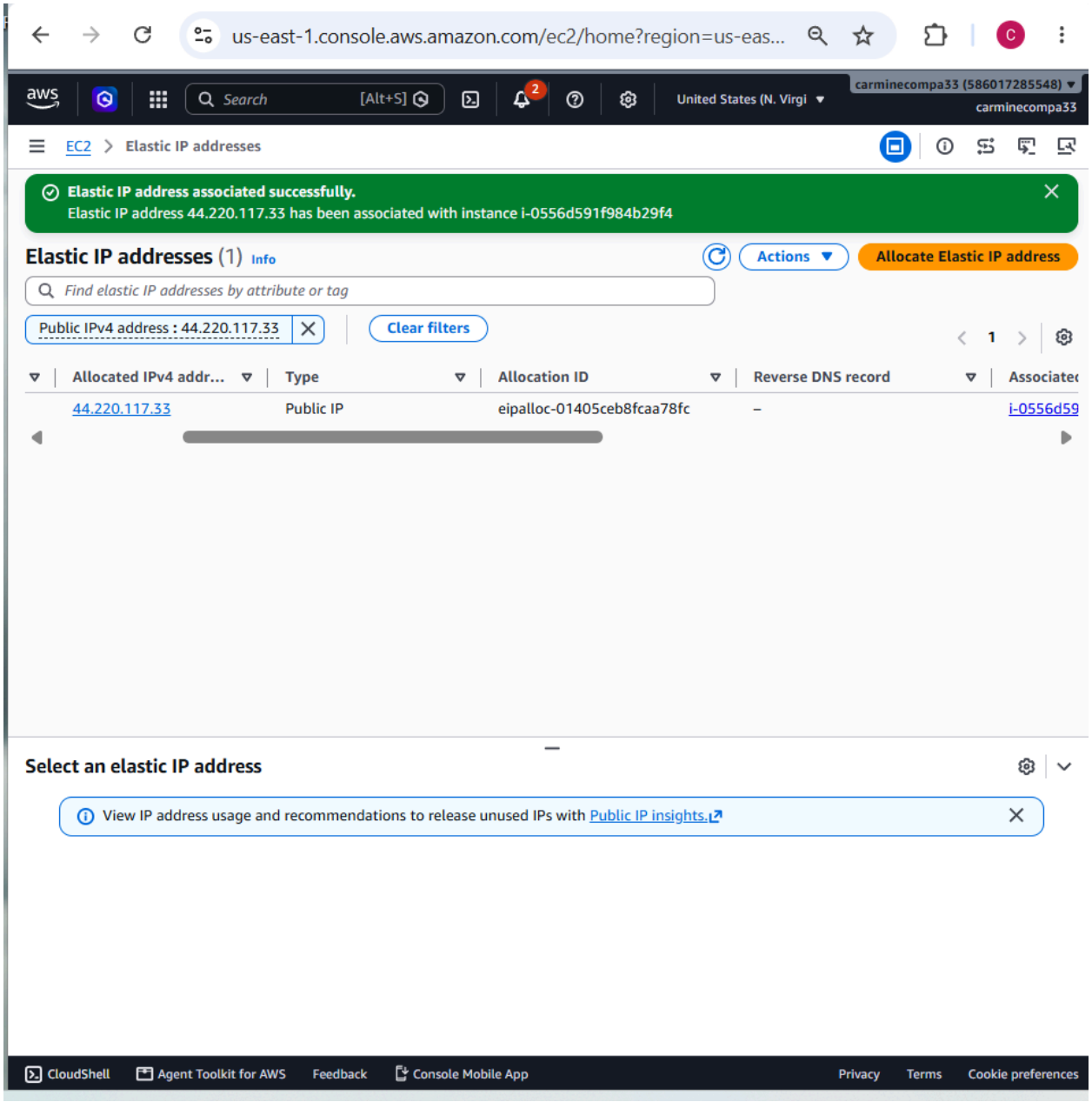



us-east-1.console.aws.amazon.com/ec2/home?region=us-eas...
Search
[Alt+S]
United States (N. Virgi
carminecompa33 (586017285548)
carminecompa33
EC2 > Security Groups > sg-068727310daa9ba09
Outbound security group rules successfully modified on security group (sg-068727310daa9ba09 | launch-wizard-1)
Details
sg-068727310daa9ba09 - launch-wizard-1
Actions
Details
Security group name
launch-wizard-1
Security group ID
sg-068727310daa9ba09
Description
launch-wizard-1 created 2026-09-14T13:00:29.146Z
VPC ID
vpc-00af755232f312767
Owner
586017285548
Inbound rules count
4 Permission entries
Outbound rules count
4 Permission entries
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (4)
Manage tags
Edit inbound rules
Search
1
Type
Protocol
Port range
Source
Description
ESP (50)
ESP (50)
All
0.0.0.0/0
IP Protocol
SSH
TCP
22
0.0.0.0/0
–
Custom UDP
UDP
500
0.0.0.0/0
IKE
Custom UDP
UDP
4500
0.0.0.0/0
IPsec NAT-T
CloudShell
Feedback
Privacy
Terms
Cookie preferences

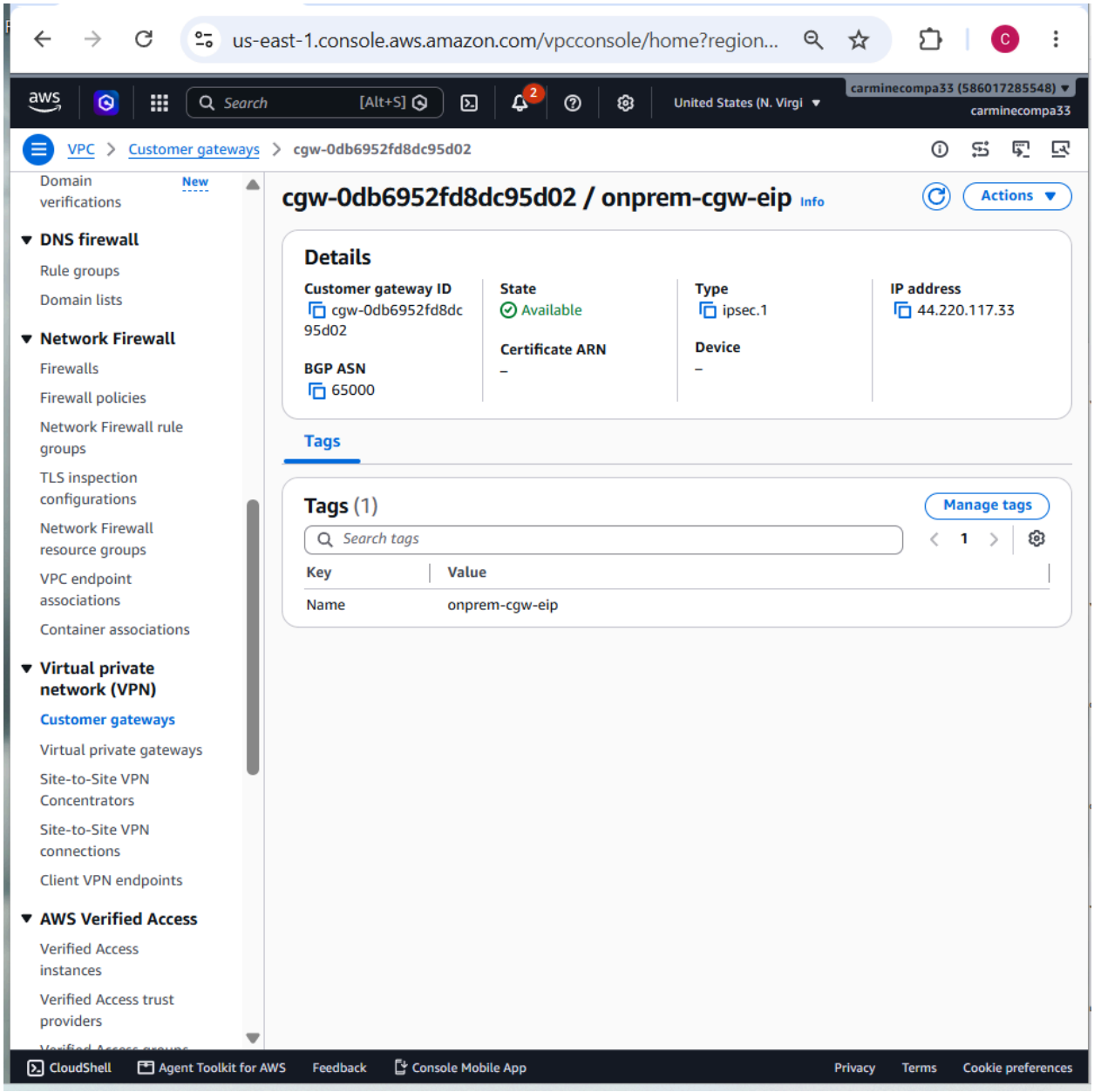
Now “sudo ipsec status” shows “ Total IPsec connections: loaded 1, active 0”



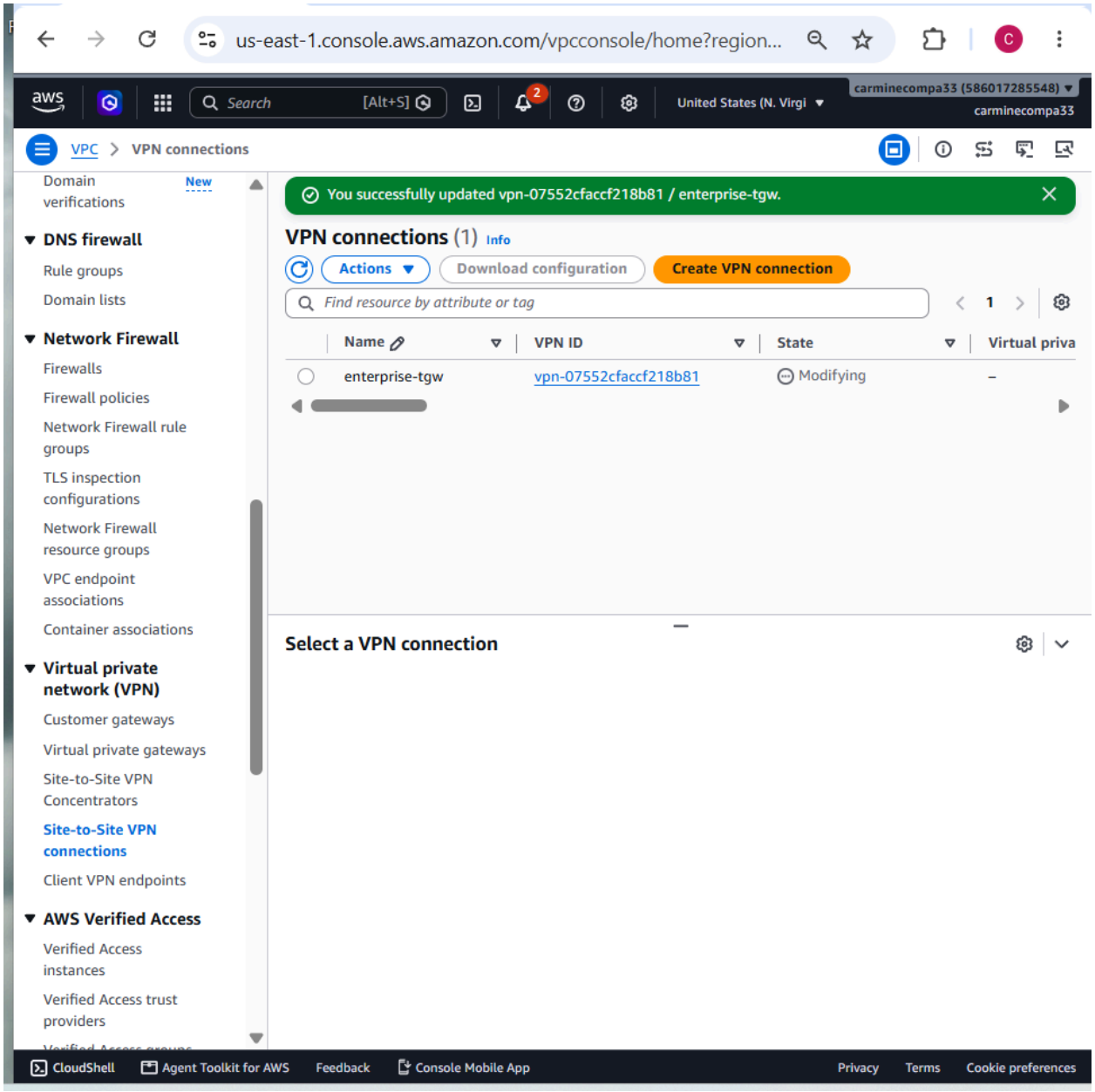
I noticed that EC2 IP public IP changed from the last time
I attach an Elastic IP to the EC2



Created a new customer gateway with the new Elastic IP address

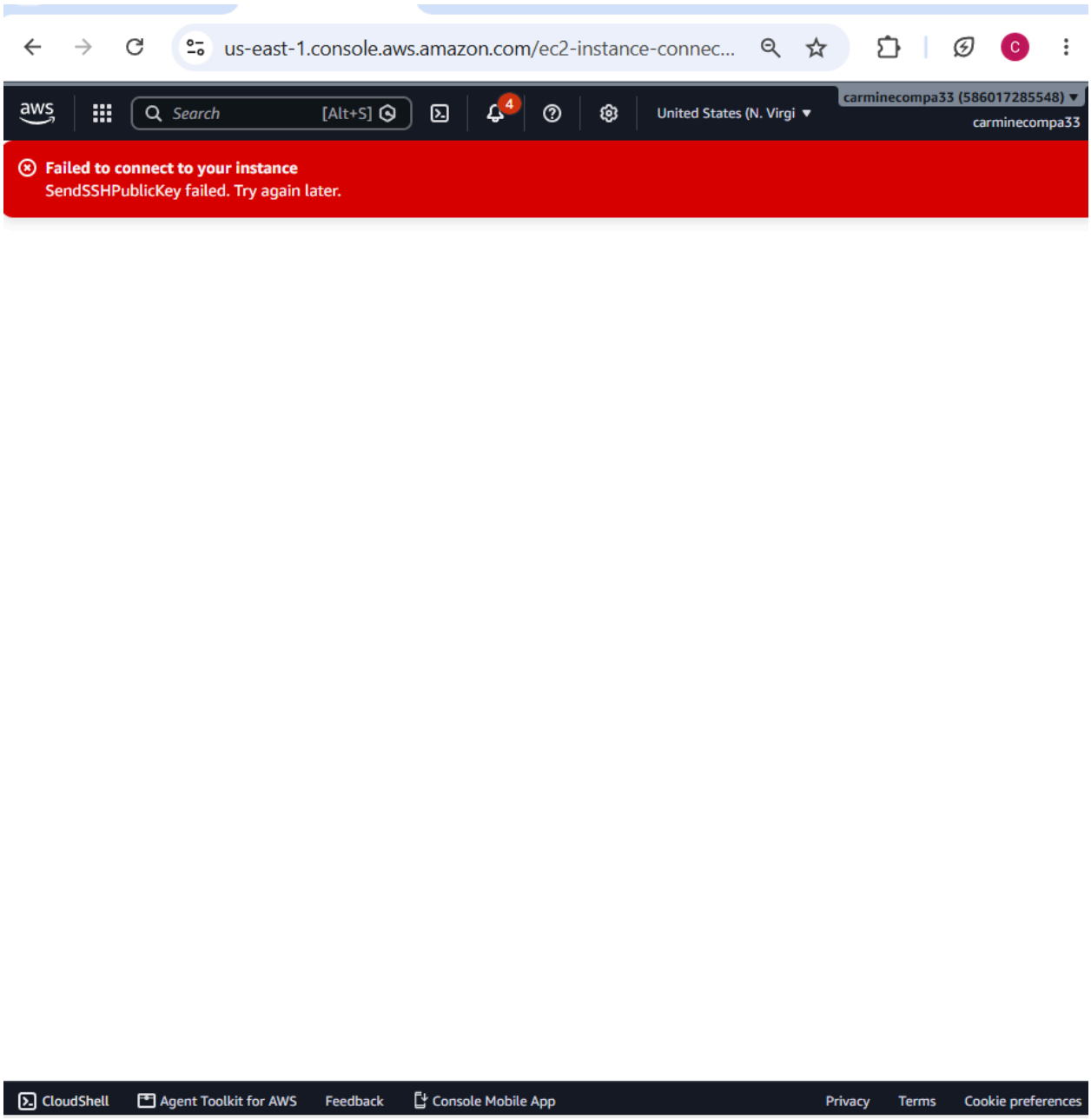


I attach the Elastic IP address to the Site-to-Site VPN connection



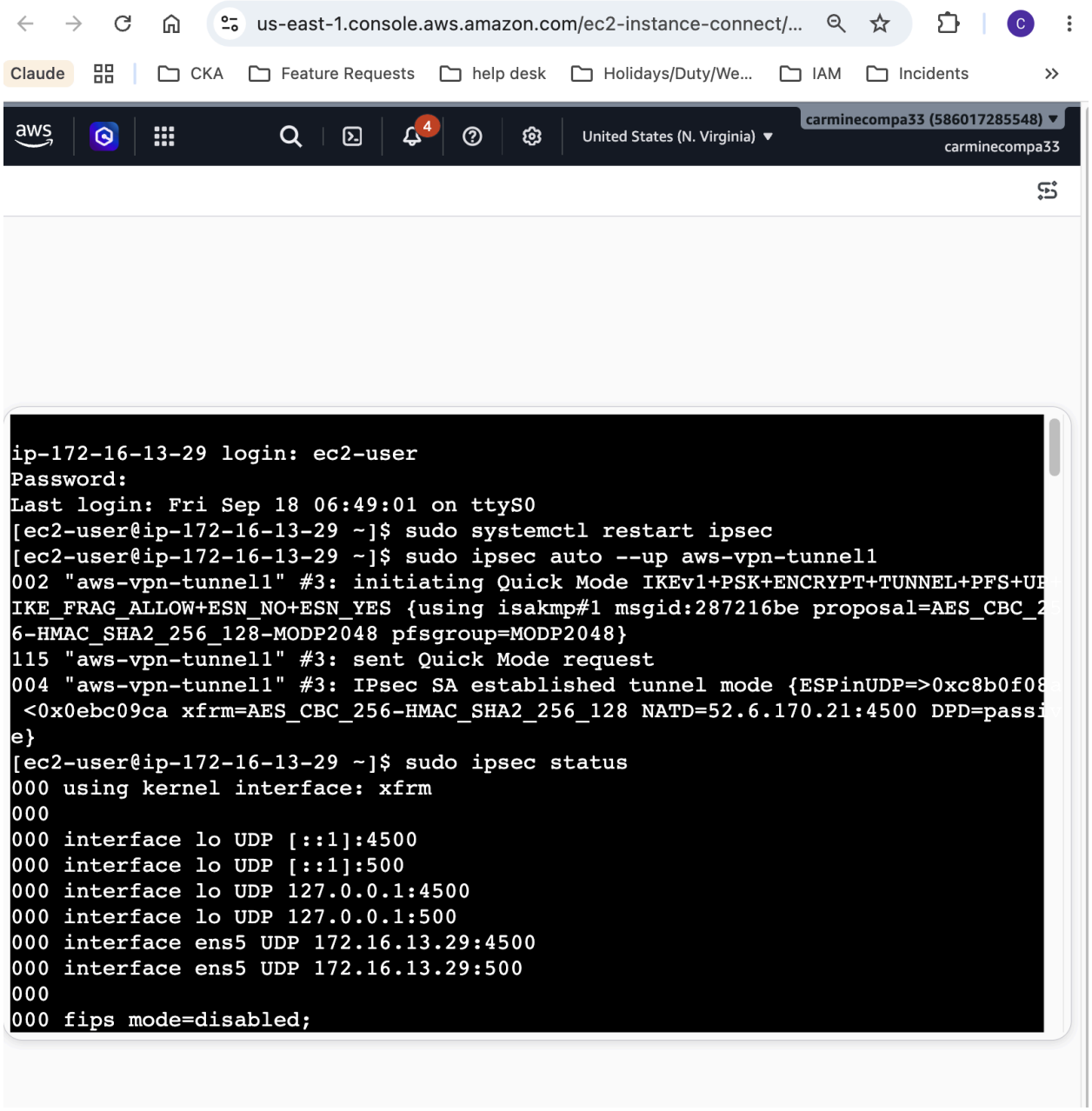
Suddenly EC2 does not allow SSH anymore to SSH. Something changed INSIDE the VM

Nothing helps



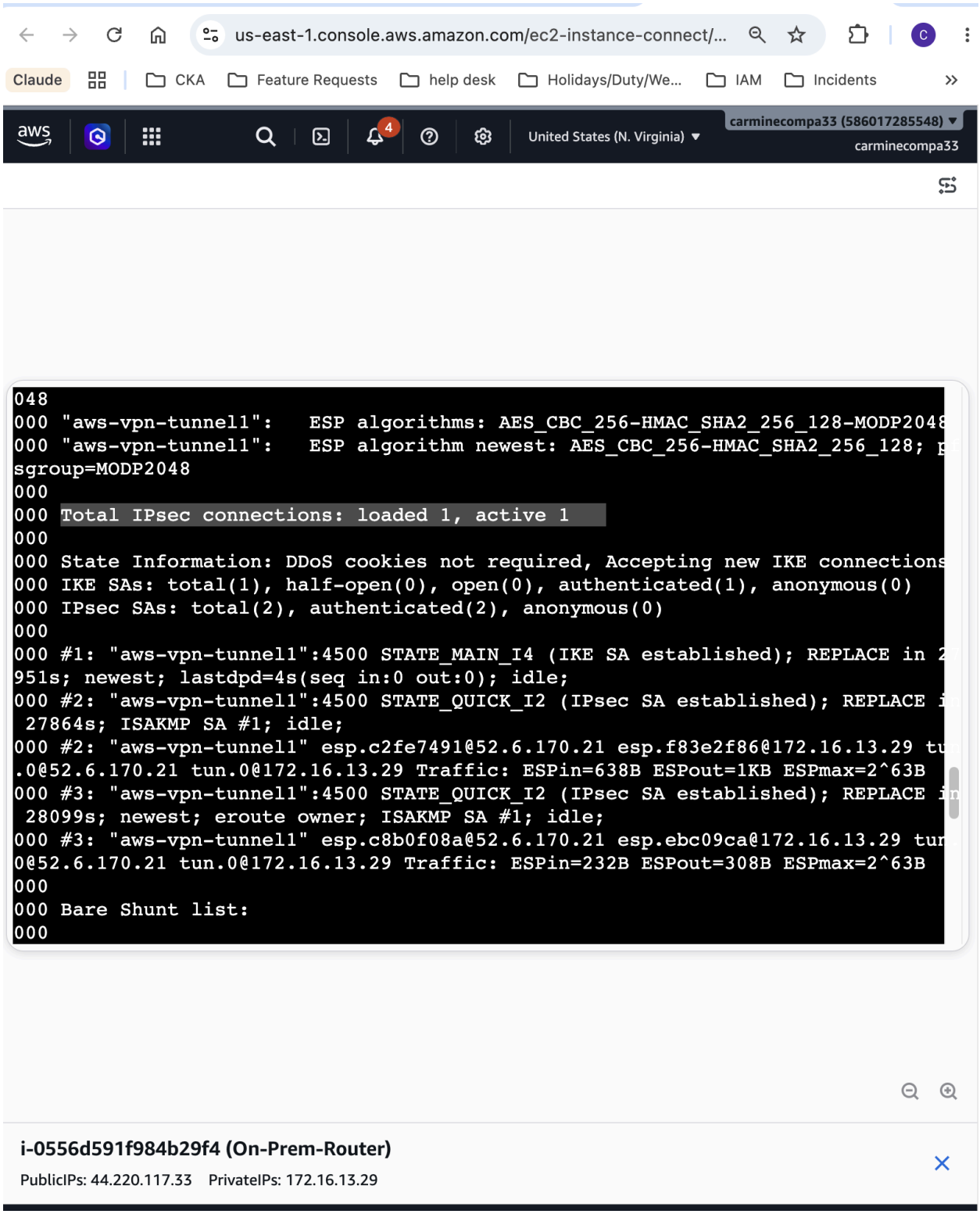
I am connecting with EC2 Serial console

And run the below commands:
sudo systemctl restart ipsec sudo ipsec auto --up aws-vpn-tunnel1

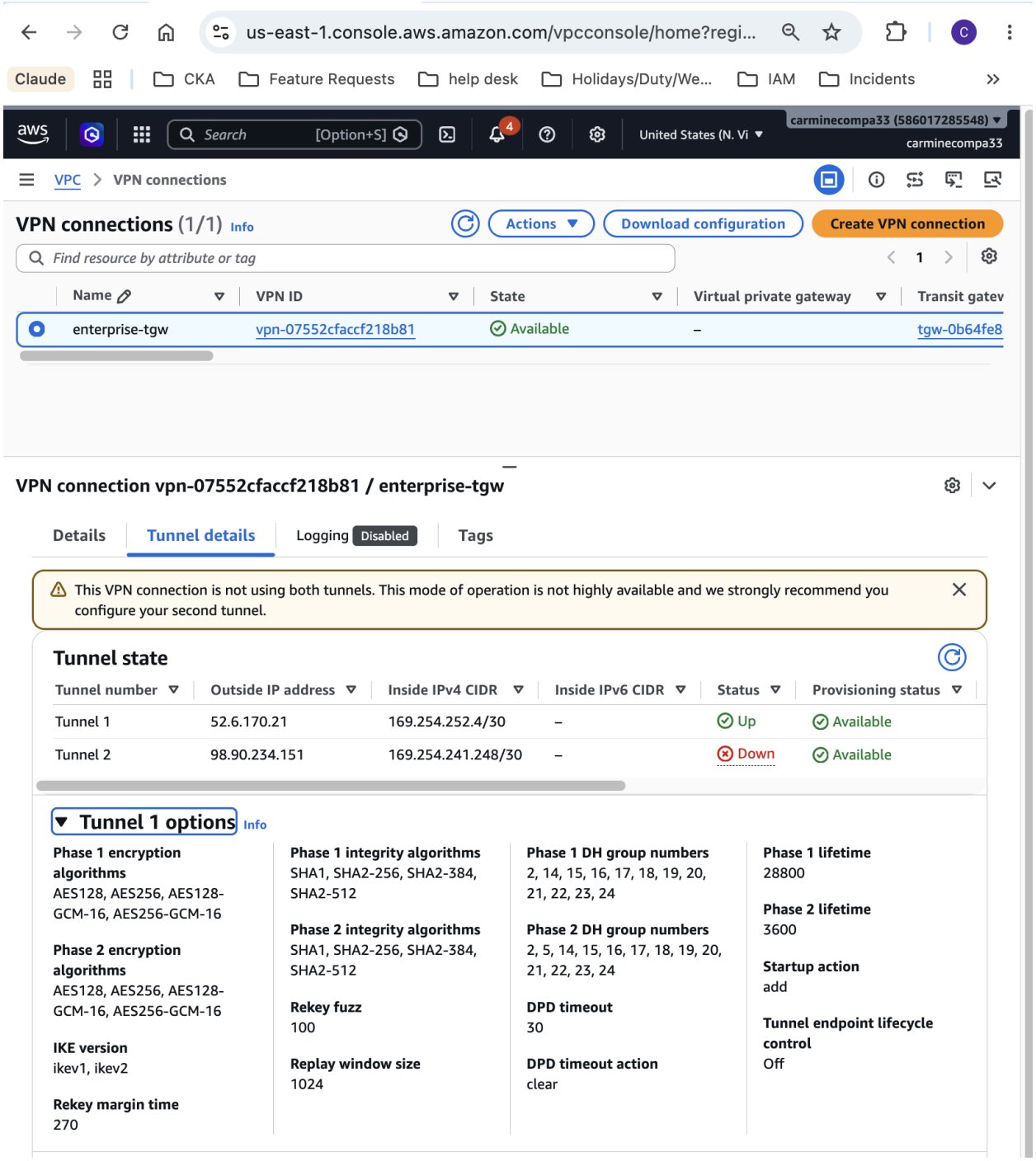


And now IpSec tunnel seem to work

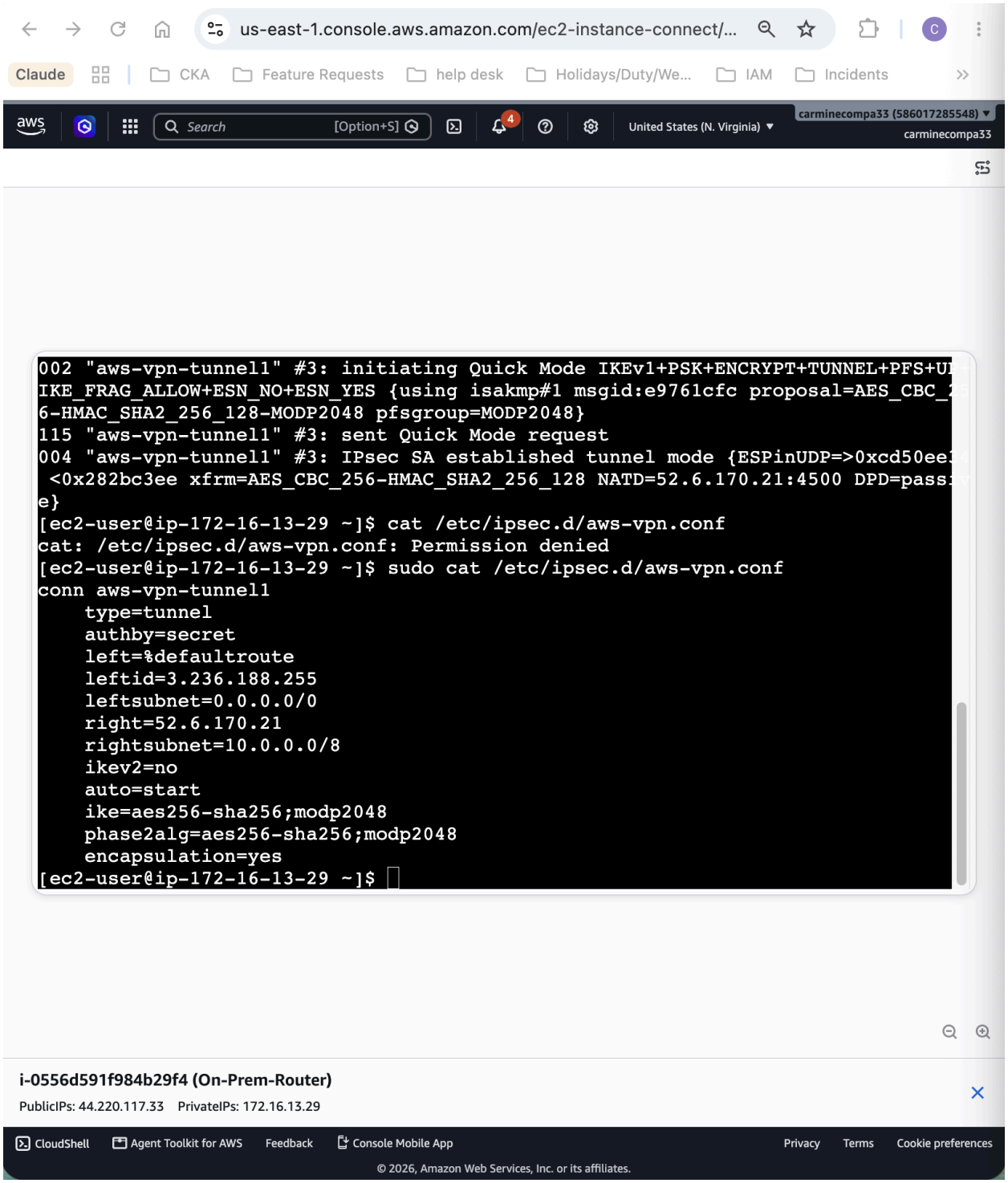
Total IPsec connections: loaded 1, active 1



I can see also from AWS in VPN site to site connection - tunnel 1



Changed right subnet from 0.0.0.0/0 to 10.0.0.0/8 >> now ssh also works



Ssh connection:

